EWB COURSES

AWS Cloud Internship
Project Submission Report

VPC Peering – Remote Access Using Bastion with Private Workloads

Submission detail	Information
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Organization	EWB Courses
Program	AWS Cloud Internship
Batch	No. 1
AWS Region	Confirm from console: ap-south-1 (Mumbai) or eu-west-1 (Ireland)
Completion date	[21 OCT 2026]
Project status	Completed

Prepared for technical review and reproducibility

Document Control

Field	Value
Report title	VPC Peering – Remote Access Using Bastion with Private Workloads
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Organization	EWB Courses
Version	1.0
Date	[21-10-2026]

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1. Project Overview and Objectives

This project demonstrates private communication between two Amazon VPCs using VPC Peering. A Bastion Host in the public subnet of VPC A provides SSH entry to private workloads. App-A and App-B communicate using private IP addresses through the peering connection. The configuration and test outcomes must match the actual AWS console and terminal evidence.

Objectives

- Create two VPCs with non-overlapping IPv4 CIDR ranges.
- Create and associate the required public/private subnets and route tables.
- Establish an active VPC peering connection and configure forward and return routes.
- Configure security group inbound rules for SSH and ICMP testing.
- Demonstrate the permitted SSH path and private-IP connectivity.
- Show a controlled ICMP failure and recovery, then document resource cleanup.

Scope and assumptions

The scope of this project is to design and implement secure private communication between two Amazon Virtual Private Clouds (VPCs) using AWS VPC Peering. VPC A contains a public subnet with a Bastion Host and a private subnet hosting App-A, while VPC B contains a private subnet hosting App-B. The Bastion Host provides SSH access to private instances, and the peering connection enables communication between App-A and App-B using their private IP addresses.

The project includes configuring VPCs, subnets, route tables, route associations, VPC peering, and security group rules. Connectivity is validated through SSH and ICMP (ping) tests, including a controlled failure and recovery scenario. Finally, the project documents the deployed resources, test results, and cleanup activities.

- Both VPCs use non-overlapping IPv4 CIDR ranges to enable private communication through VPC Peering.
- The VPC peering connection is active, and the required forward and return routes are configured correctly.
- Private instances do not have public IP addresses and are accessed through the Bastion Host or permitted private network paths.
- Security groups allow only the required SSH and ICMP traffic for administration and connectivity testing.
- VPC Peering provides private connectivity between the VPCs but does not automatically provide internet access or transitive routing.
- All resource IDs, private IP addresses, route targets, and connectivity results are verified against the actual AWS console.
- Temporary security group changes made during testing are reverted after the tests are completed.

2. Architecture Diagram

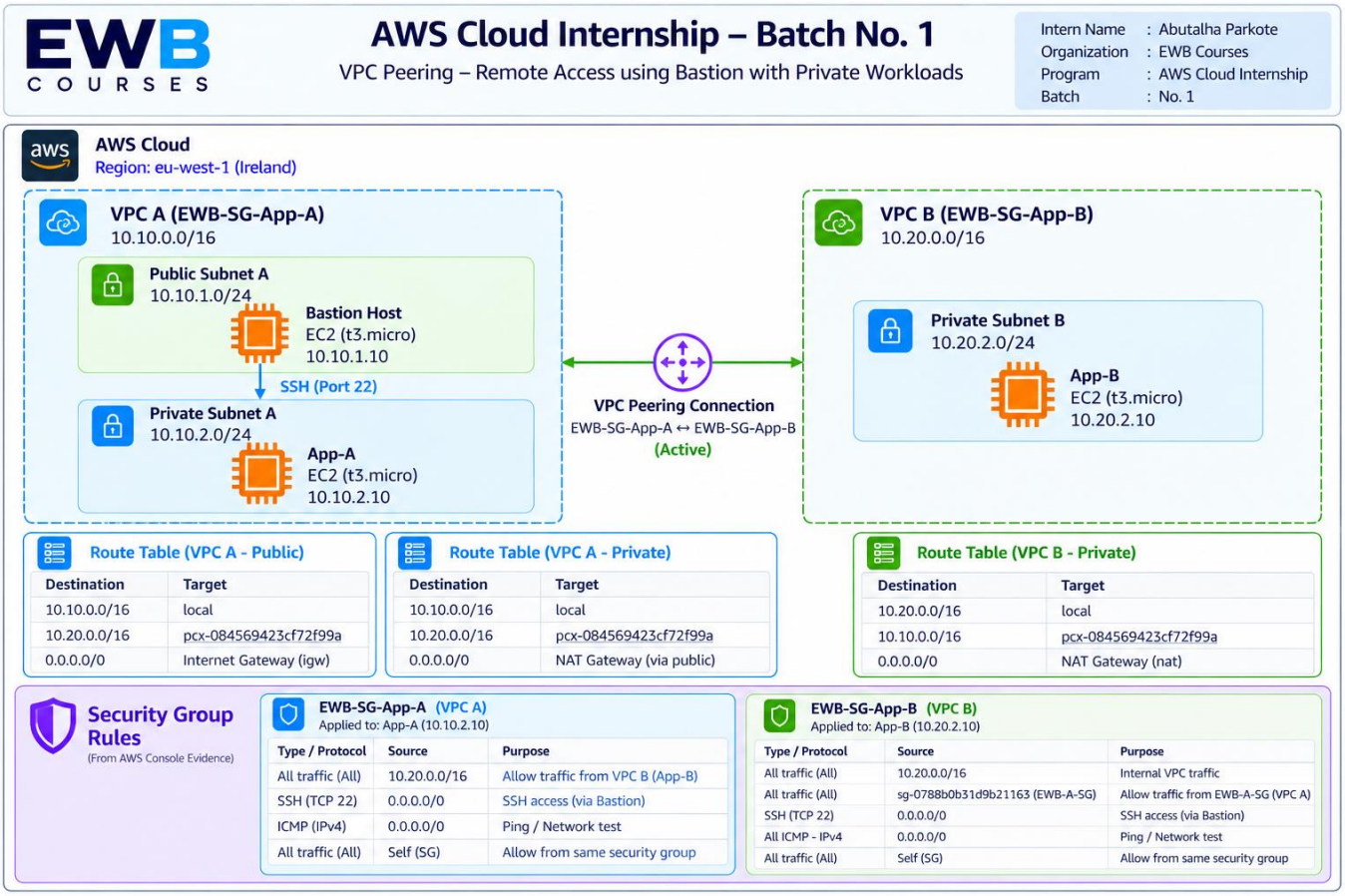
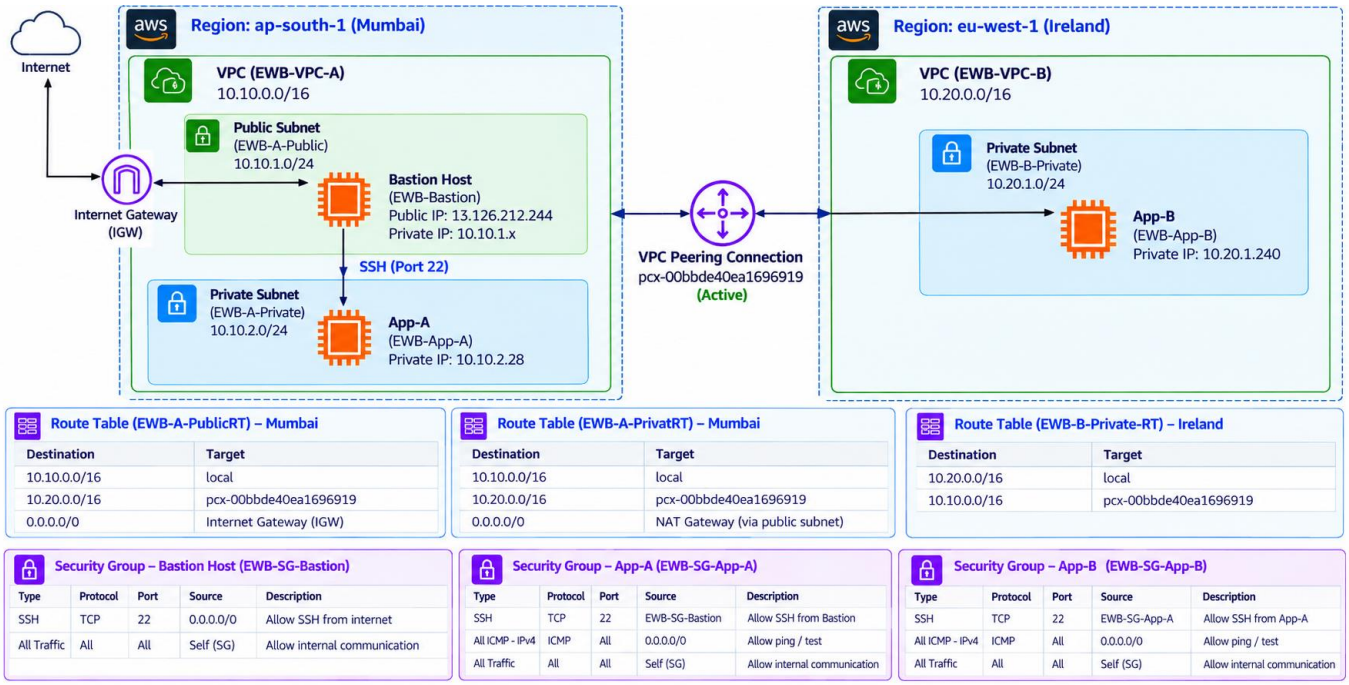


Figure 1. Final VPC peering architecture (label all resource IDs and private IPs where available).



3. Resource Inventory

Fill every field from the AWS console. Do not include private keys, passwords, tokens, or credentials.

Resource	Name / ID	Region	CIDR / Private IP	Purpose / Status
VPC A	EWB-VPC-A	Mumbai	10.10.0.0/16 (verify)	VPC containing Bastion and App-A
VPC B	EWB-VPC-B	Ireland	10.20.0.0/16 (verify)	VPC containing App-B
Public subnet A	EWB-Public-A	Mumbai	10.10.1.0/24 (verify)	Bastion subnet
Private subnet A	EWB-Private-A	Mumbai	10.10.2.0/24 (verify)	App-A subnet
Private subnet B	Ewb-Private-B	MUMBAI	10.20.2.0/24 (verify)	App-B subnet
Bastion EC2	EWB-Bastion	Mumbai - a	10.10.1.10 (verify)	SSH entry point
App-A EC2	EWB-App-A	Mumbai - b	10.10.2.10 (verify)	Private workload
App-B EC2	EWB-App-B	Ireland - a	10.20.2.10 (verify)	Private workload
VPC peering	EWB-A-B-Peer	[Both regions]	10.10.0.0/16 ↔ 10.20.0.0/16	[Active / other status]

4. VPC and Subnet Configuration

VPC	IPv4 CIDR	Subnet	Subnet CIDR	AZ	Route table association
VPC A	[Confirm]	Public subnet A	[Confirm]	[Confirm]	[Public RT ID/name]
VPC A	[Confirm]	Private subnet A	[Confirm]	[Confirm]	[Private RT ID/name]
VPC B	[Confirm]	Private subnet B	[Confirm]	[Confirm]	[Private RT ID/name]

[Add add VPC and subnet console screenshots]

5. Route Tables and Associations

Document every route table and its subnet association. Include the local route, peering route, and default route only when configured.

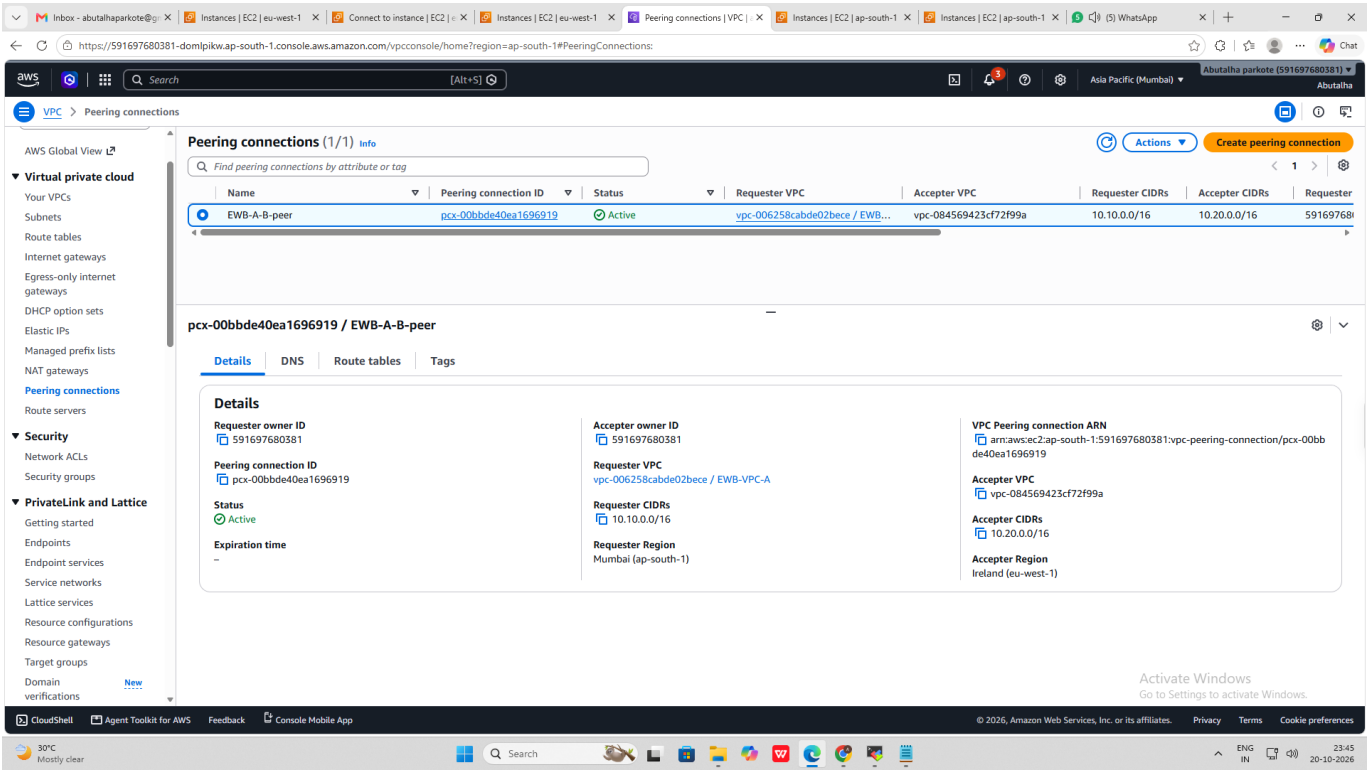
VPC / Route table	Associated subnet(s)	Destination CIDR	Target	Status / notes

VPC A – Public RT	[Subnet ID]	10.10.0.0/16	local	[Active]
VPC A – Public RT	[Subnet ID]	10.20.0.0/16	[pcx- ID]	[Active]
VPC A – Public RT	[Subnet ID]	0.0.0.0/0	[Internet Gateway ID]	[Verify]
VPC A – Private RT	[Subnet ID]	10.10.0.0/16	local	[Active]
VPC A – Private RT	[Subnet ID]	10.20.0.0/16	[pcx- ID]	[Active]
VPC A – Private RT	[Subnet ID]	0.0.0.0/0	[NAT target only if configured]	[Verify / omit if absent]
VPC B – Private RT	[Subnet ID]	10.20.0.0/16	local	[Active]
VPC B – Private RT	[Subnet ID]	10.10.0.0/16	[pcx- ID]	[Active]
VPC B – Private RT	[Subnet ID]	0.0.0.0/0	[Actual target, if present]	[Verify / omit if absent]

[Add add route table screenshots with Routes and Subnet associations tabs visible]

6. VPC Peering Configuration

Field	Value
Peering connection name	EWB-A-B-Peer
Peering connection ID	pcx- 00bbde40ea1696919
Requester VPC / CIDR	VPC-084569423cf72f99a/ 10.10.0.0/16
Accepter VPC / CIDR	VPC-00bbde40ea1696919 / 10.20.0.0/16
Requester Region	Mumbai
Accepter Region	Ireland
Connection status	[Active — verify in console]

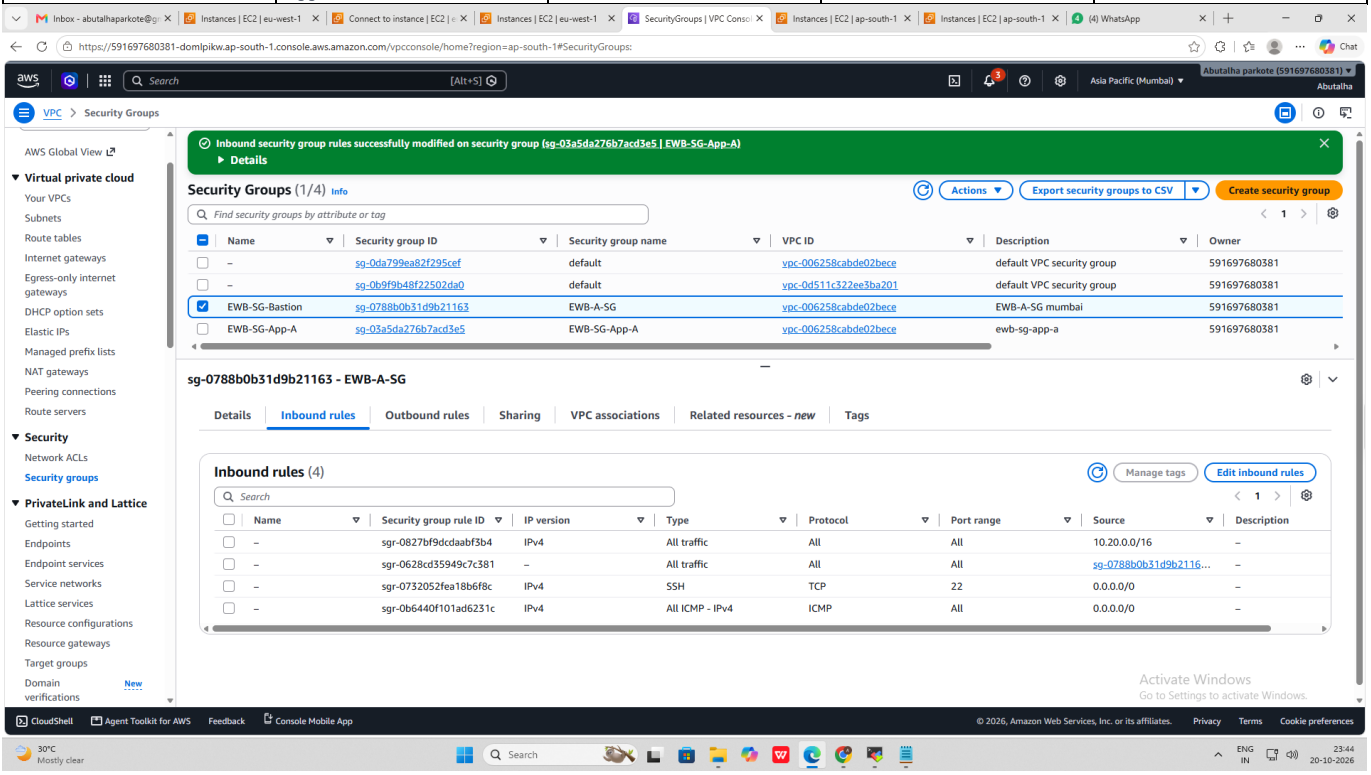


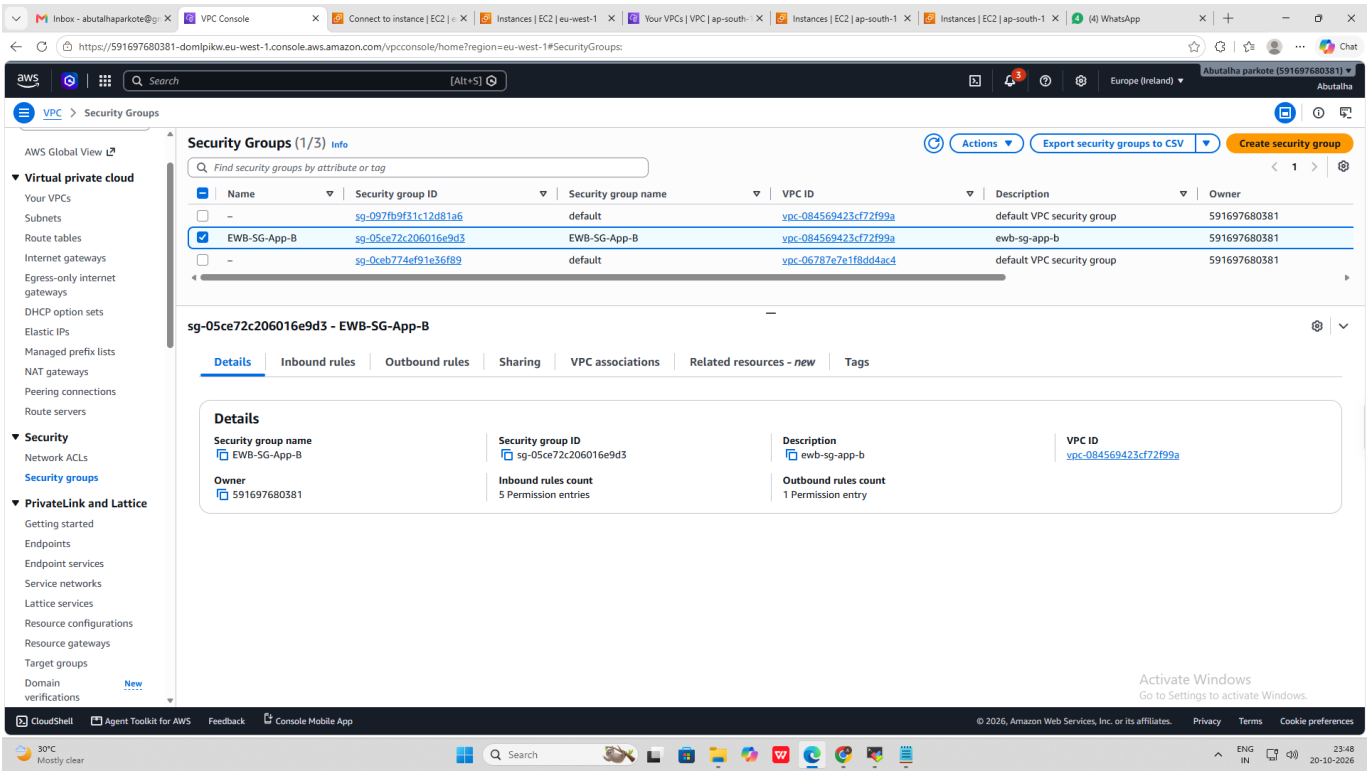
7. Security Group Rules

Record each inbound rule exactly as shown in the AWS console, including protocol, port range, source, and description.

Security group / attached resource	Type / protocol	Port range	Source	Reason

Bastion SG / Bastion	SSH / TCP	22	[Your trusted public IP/32]	Administrative SSH access
App-A SG / App-A	SSH / TCP	22	[Bastion SG ID or Bastion private IP/32]	SSH from Bastion
App-A SG / App-A	ICMP or required application rule	[As configured]	[As configured]	Connectivity test
App-B SG / App-B	SSH / TCP	22	[As configured]	SSH test via approved path
App-B SG / App-B	ICMP or required application rule	[As configured]	[As configured]	Connectivity test





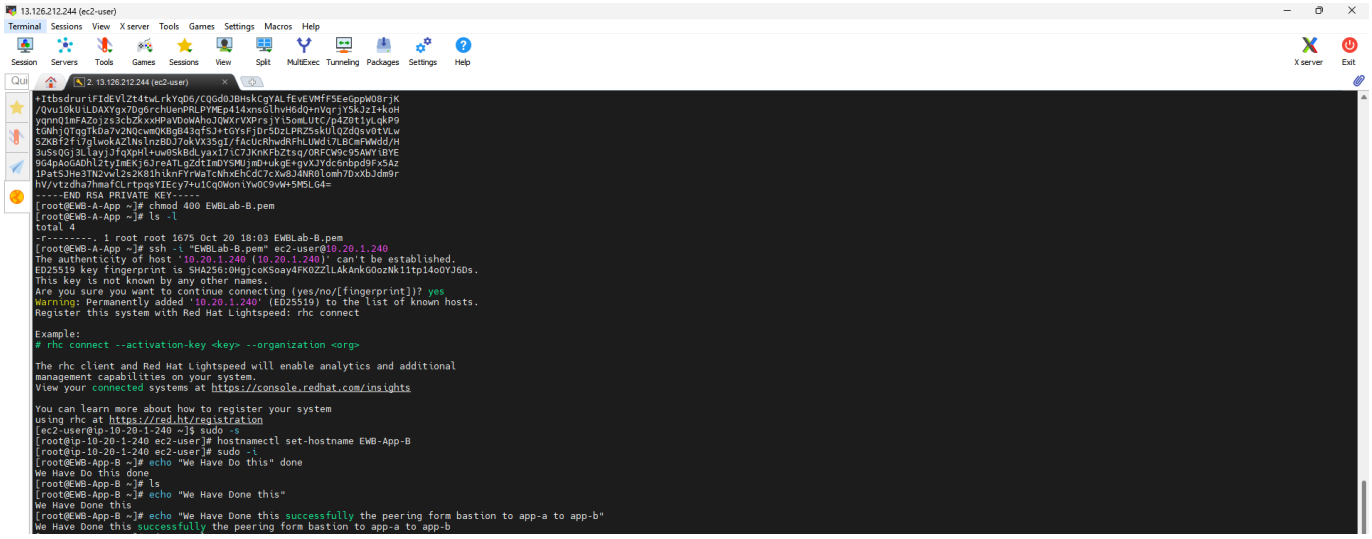
Security note: avoid SSH from 0.0.0.0/0 in a final configuration. Restrict it to a trusted IP or a suitable security-group source where supported.

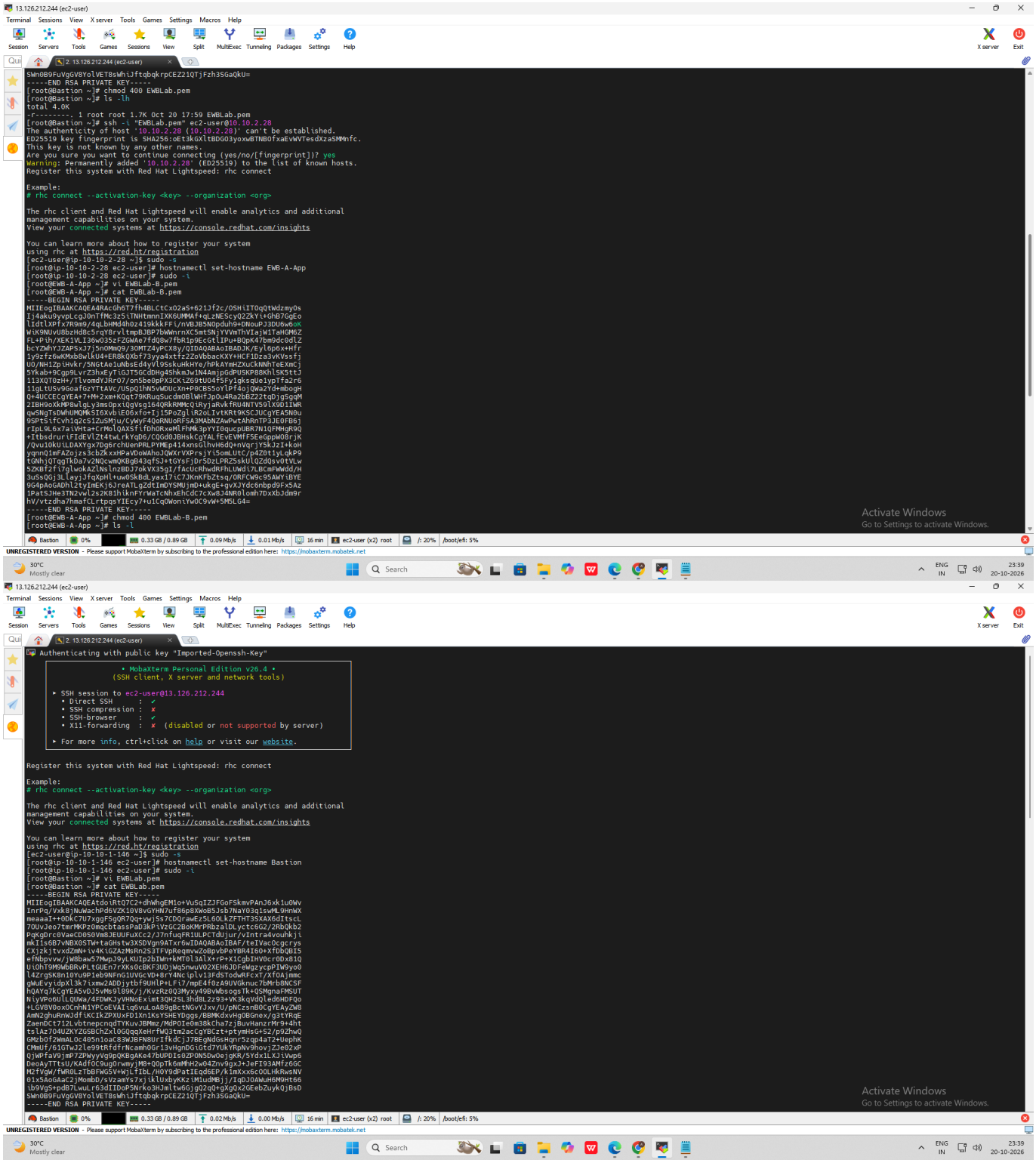
8. SSH Access and Connectivity Tests

Use private destination IPs for private instances. Include command output and a short interpretation for every test.

Test ID	Source → destination	Command / method	Expected result	Actual result / evidence
T1	Laptop → Bastion public IP	ssh -i <key.pem> ec2-user@<bastion-public-ip>	SSH succeeds	[Pass/Fail + screenshot]
T2	Bastion → App-A private IP	ssh -i <key.pem> ec2-user@10.10.2.10	SSH succeeds if SG/routes allow	[Pass/Fail + screenshot]
T3	App-A → App-B private IP	ping -c 4 10.20.2.10	Replies if ICMP and routes allow	[Pass/Fail + screenshot]
T4	App-B → App-A private IP	ping -c 4 10.10.2.10	Replies if ICMP and routes allow	[Pass/Fail + screenshot]
T5	Bastion → App-B private IP	ssh -J ec2-user@<bastion-public-ip> ec2-user@10.20.2.10	SSH succeeds if configured and permitted	[Pass/Fail + screenshot]

Use ProxyJump only if t





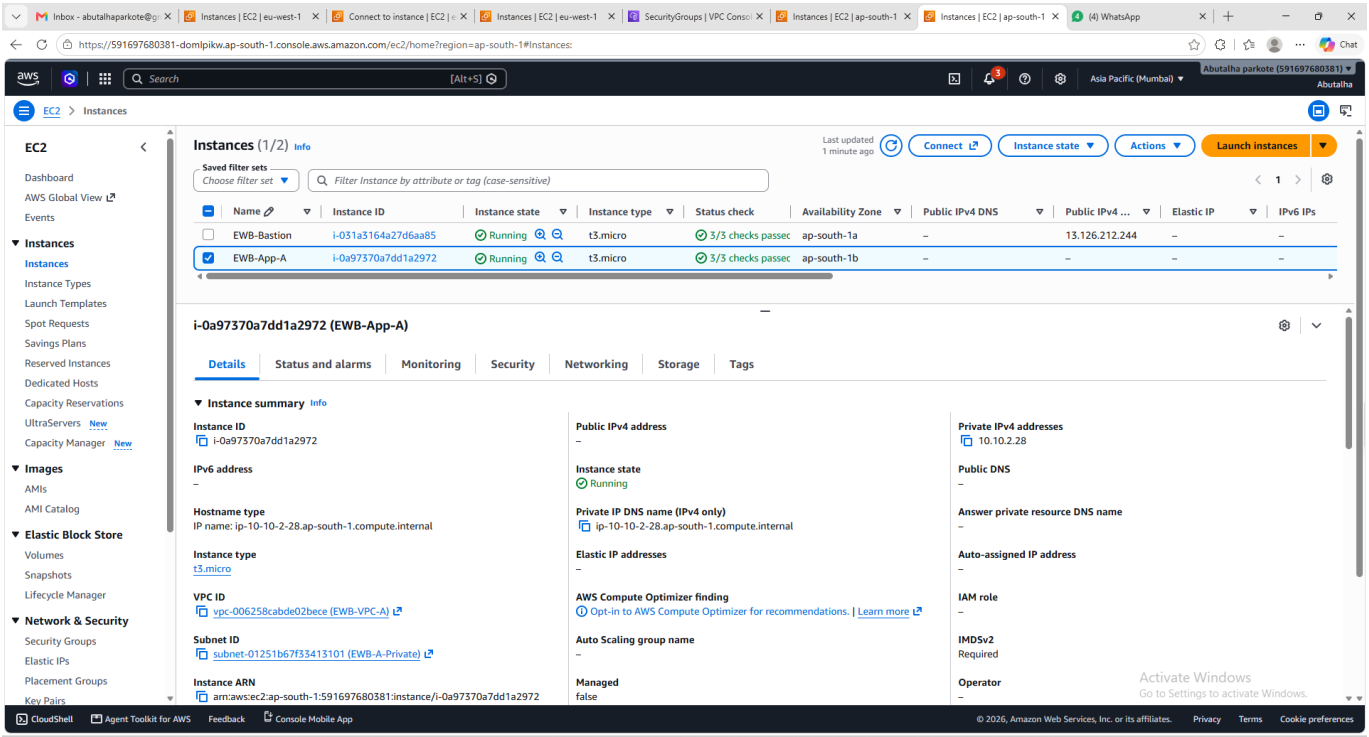
he required private key and SSH configuration are safely available. Never paste a private key into the report.

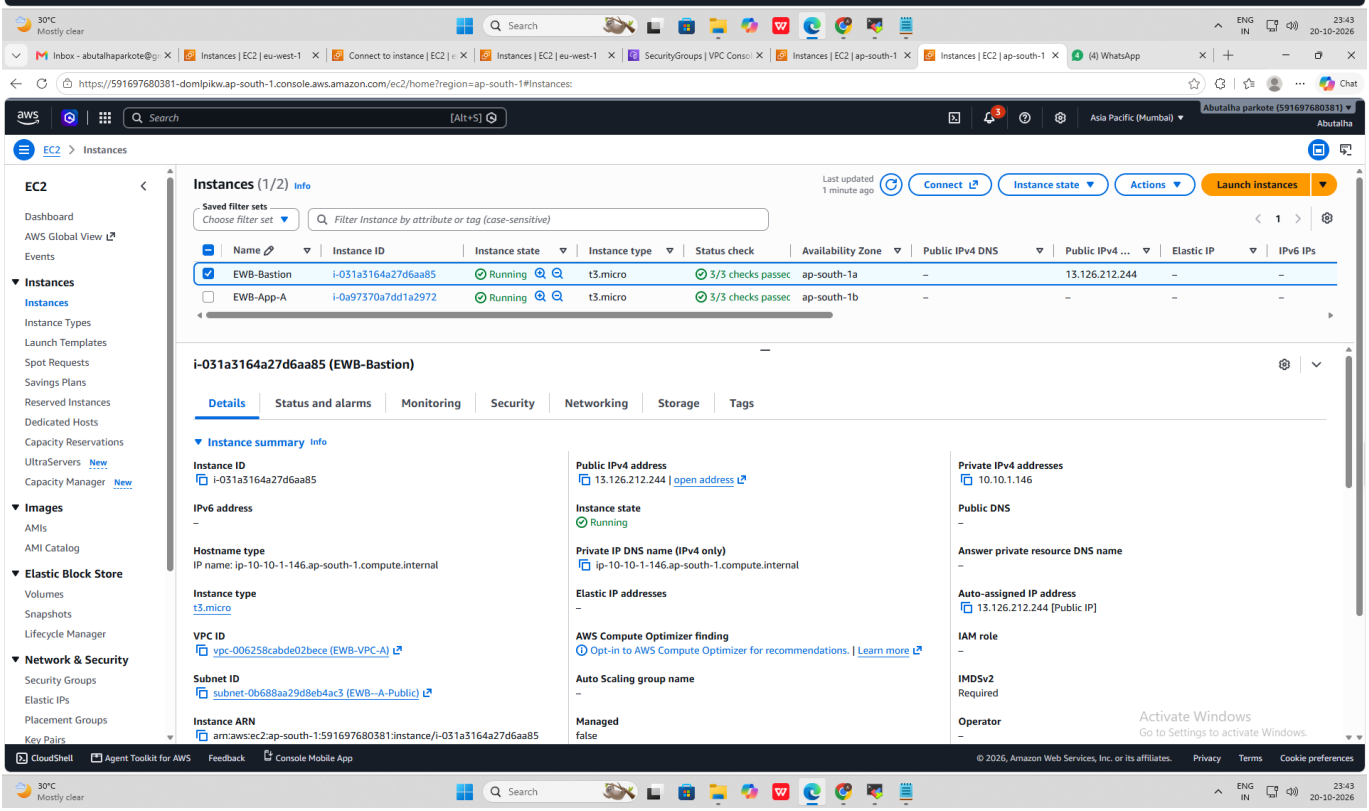
9. Cleanup Evidence

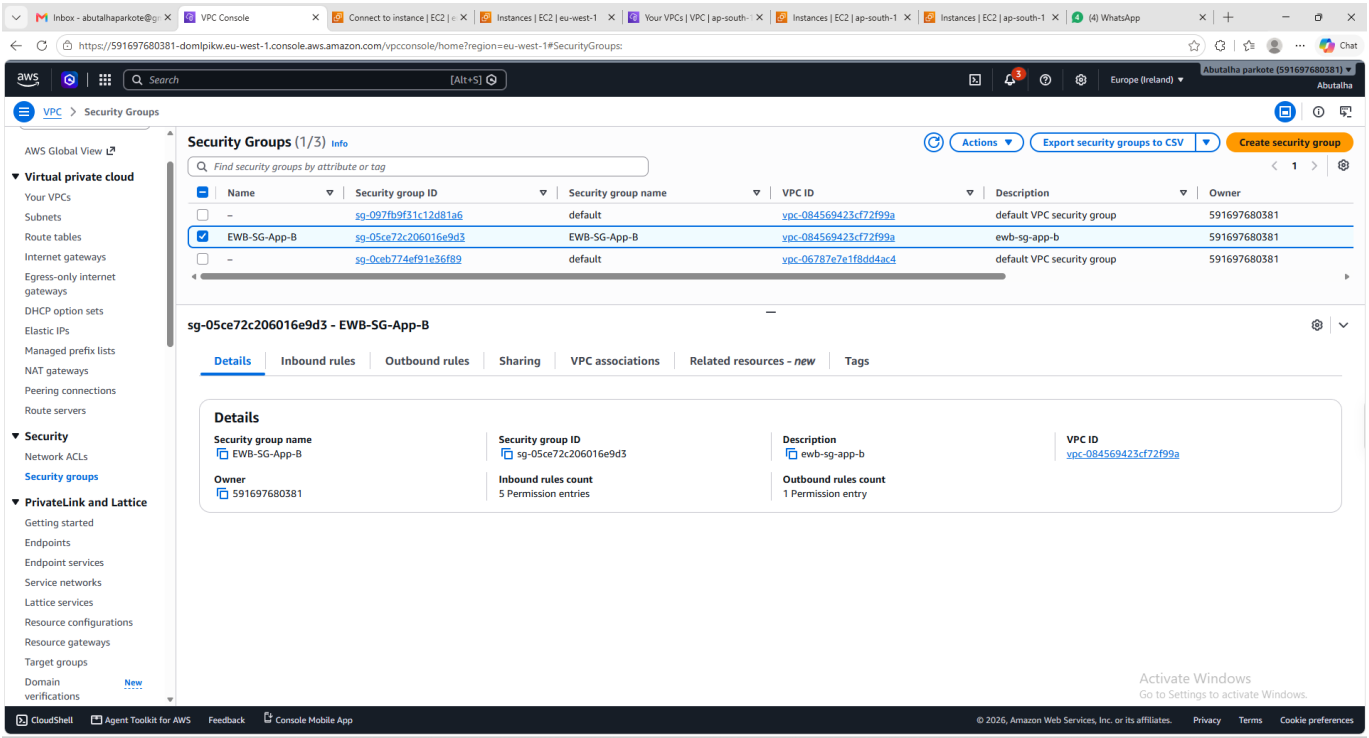
List the resources actually deleted or retained. Do not claim cleanup of a resource unless the console confirms it.

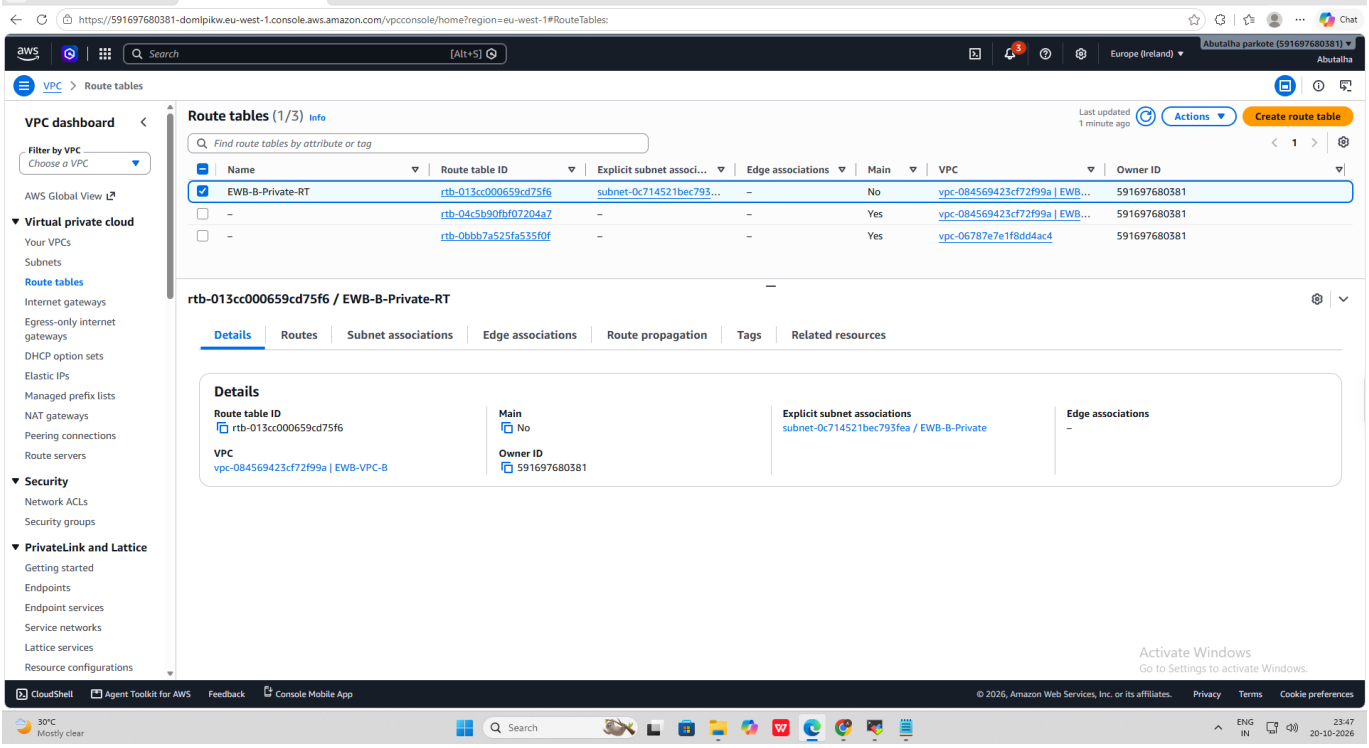
Resource	Resource ID / name	Action	Evidence / final state

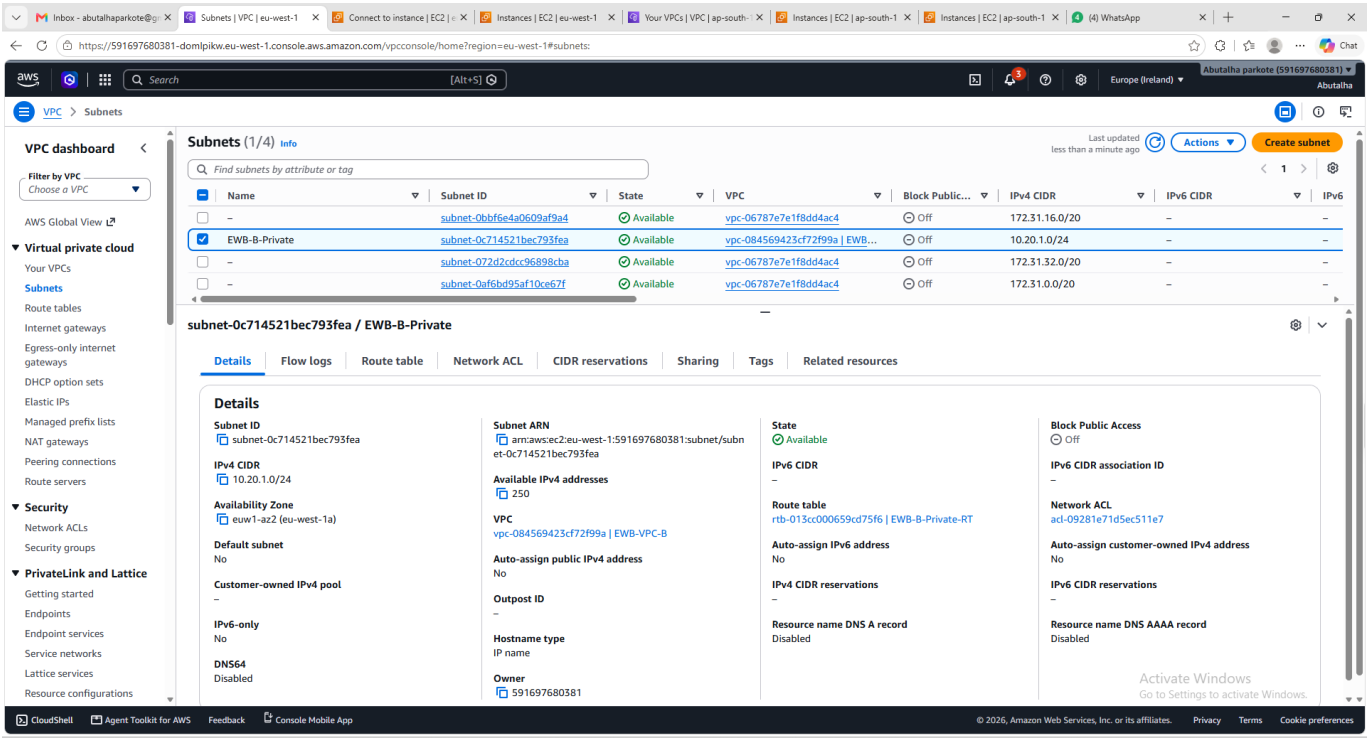
EC2 instances		[Stopped / terminated / retained]	[Screenshot]
VPC peering connection	pcx- 00bbde40ea1696919	[Deleted / retained]	[Screenshot]
NAT Gateway / Elastic IP	EWB-NAT	[Deleted / retained / not created]	[Screenshot]
Internet Gateway	EWB-A-IGW	[Detached/deleted / retained]	[Screenshot]
Subnets / route tables / security groups		[Deleted / retained]	[Screenshot]

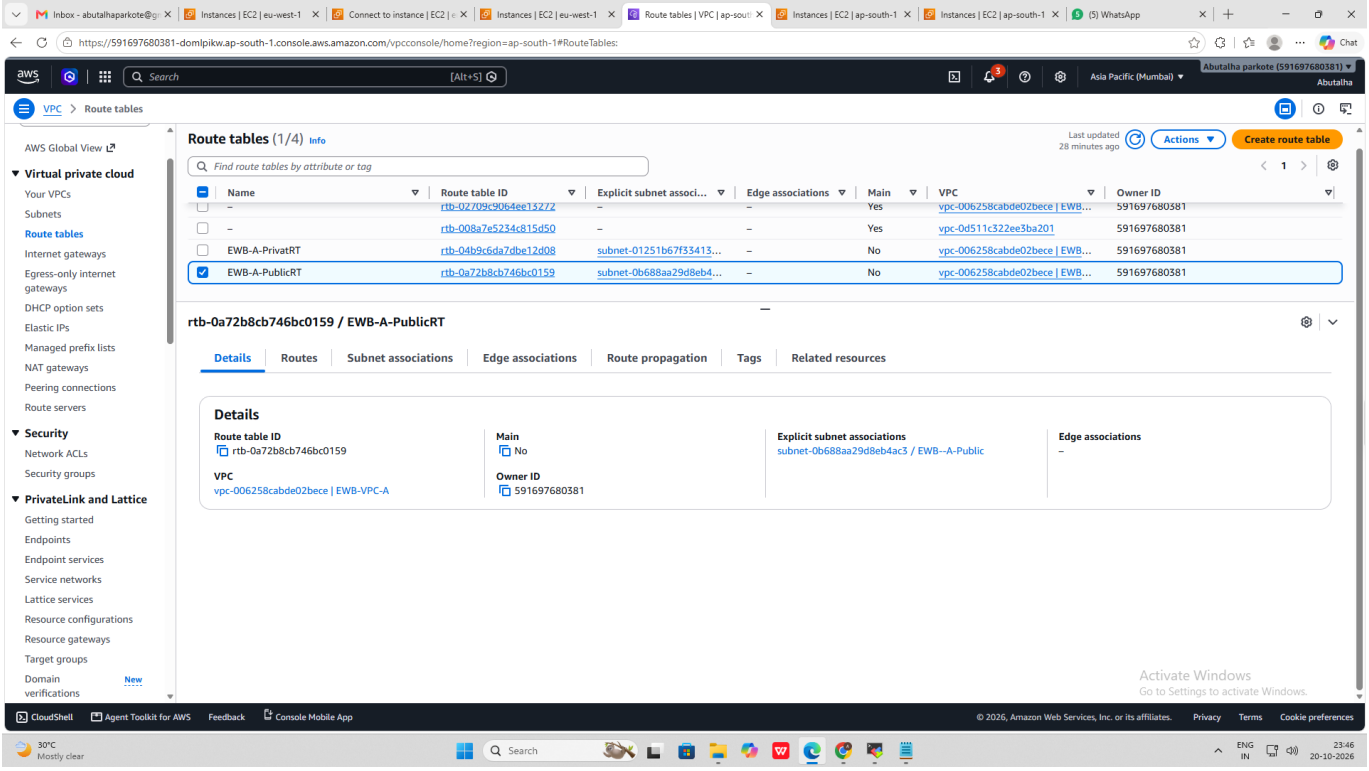


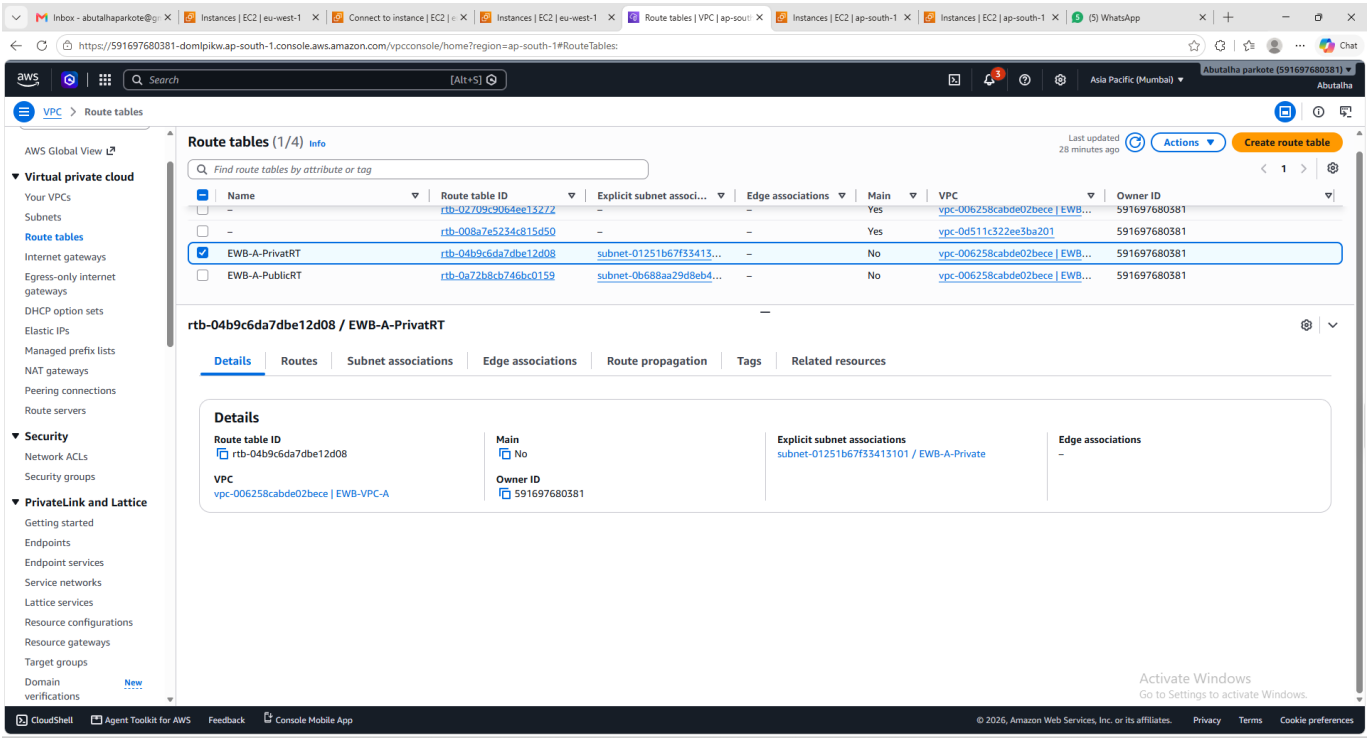


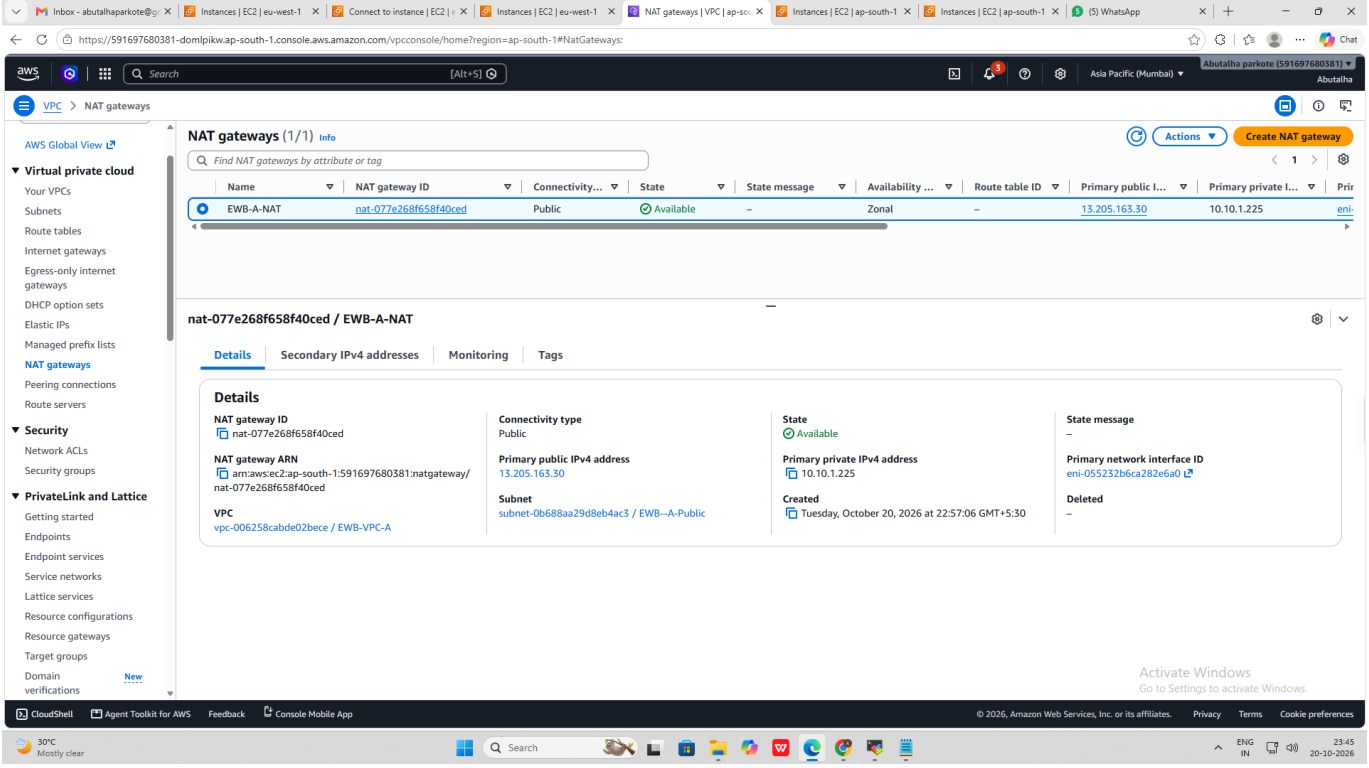

VPC Console
Security Groups (1/3) Info
Actions
Export security groups to CSV
Create security group
Name
Security group ID
Security group name
VPC ID
Description
Owner
EWB-SG-App-B
sg-05ce72c206016e9d3
EWB-SG-App-B
vpc-084569423cf72f99a
ewb-sg-app-b
591697680381
sg-05ce72c206016e9d3 - EWB-SG-App-B
Details
Inbound rules
Outbound rules
Sharing
VPC associations
Related resources - new
Tags
Security group name
EWB-SG-App-B
Security group ID
sg-05ce72c206016e9d3
Description
ewb-sg-app-b
VPC ID
vpc-084569423cf72f99a
Owner
591697680381
Inbound rules count
5 Permission entries
Outbound rules count
1 Permission entry

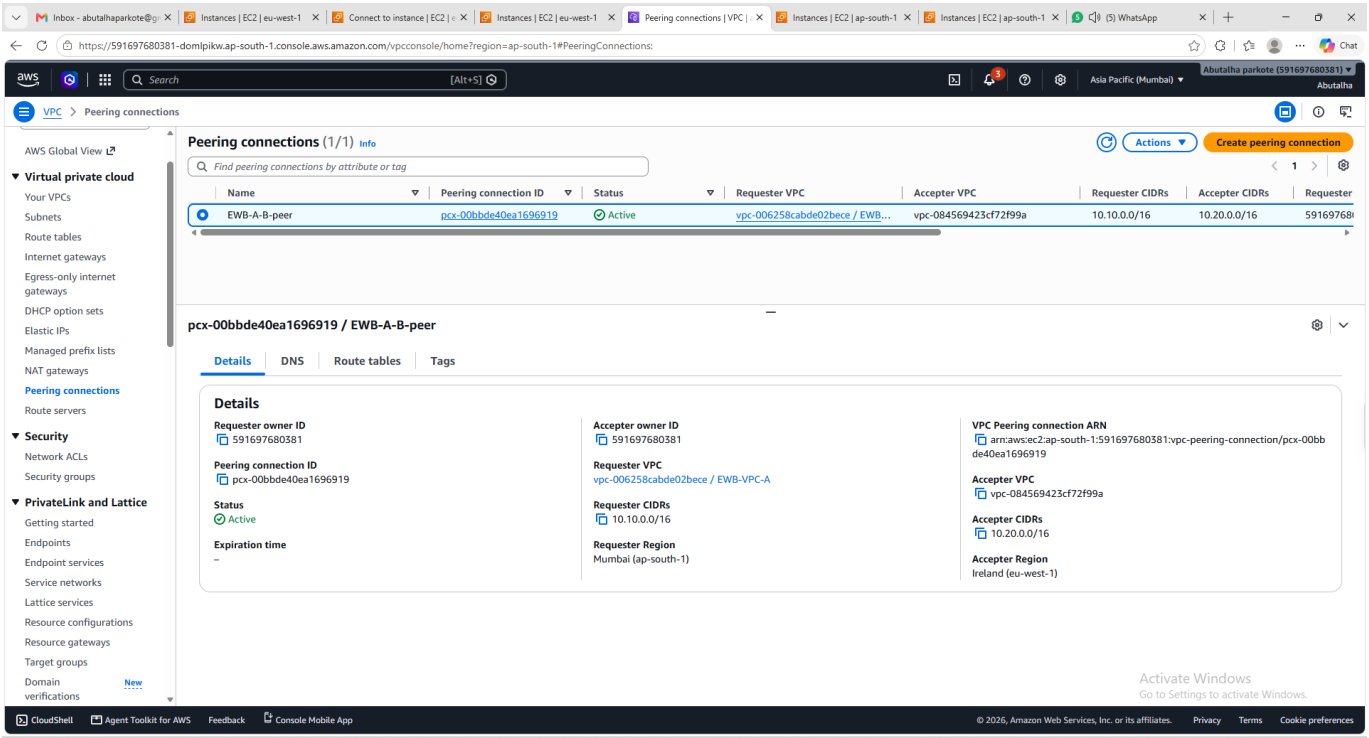

Route tables (1/3) Info
Actions
Create route table
Name
Route table ID
Explicit subnet associ...
Edge associations
Main
VPC
Owner ID
EWB-B-Private-RT
rtb-013cc000659cd75f6
subnet-0c714521bec793...
No
vpc-084569423cf72f99a | EWB...
591697680381
rtb-04c5b90fbf07204a7
Yes
rtb-0bbb7a525fa535f0f
vpc-06787e7e1f8dd4ac4
rtb-013cc000659cd75f6 / EWB-B-Private-RT
Details
Routes
Subnet associations
Edge associations
Route propagation
Tags
Related resources
Route table ID
rtb-013cc000659cd75f6
Main
No
Explicit subnet associations
subnet-0c714521bec793fea / EWB-B-Private
Edge associations
VPC
vpc-084569423cf72f99a | EWB-VPC-B
Owner ID
591697680381

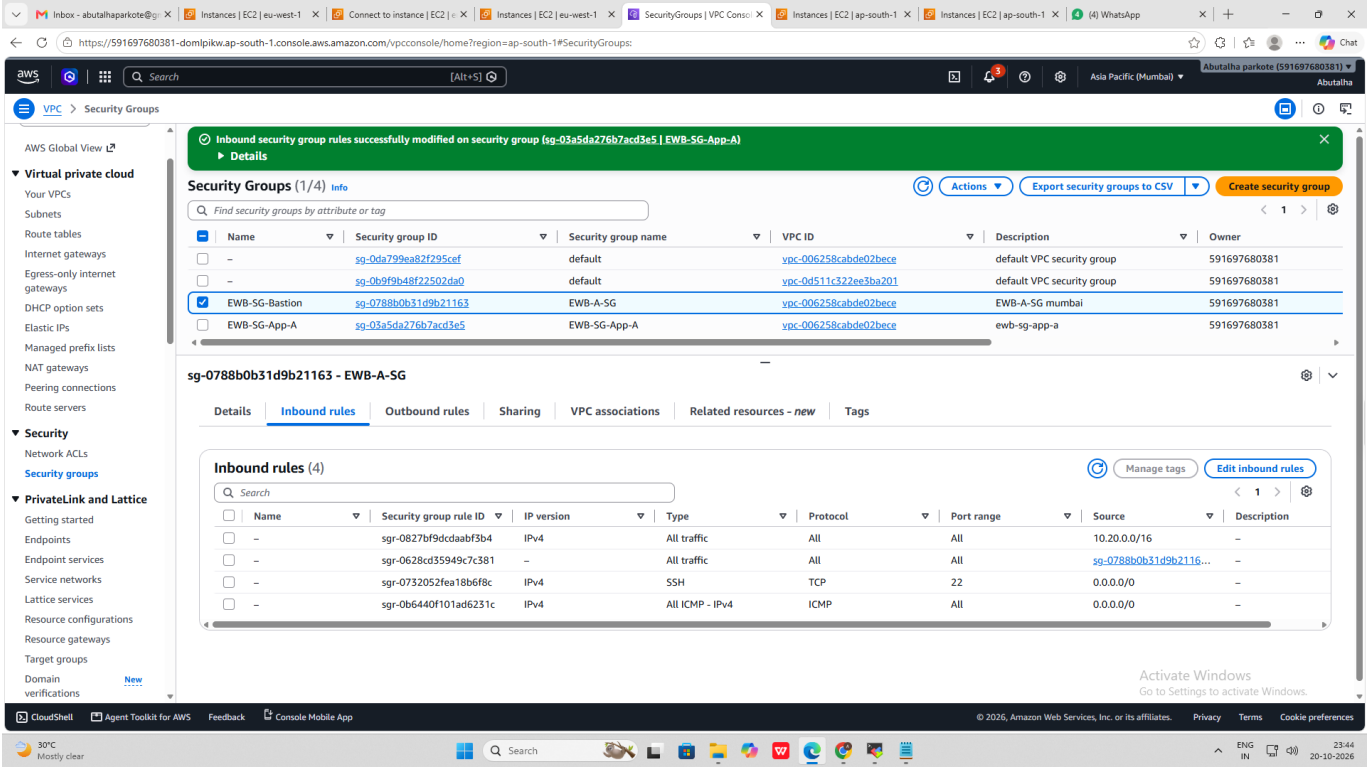

VPC dashboard
Subnets (1/4)
Create subnet
EWB-B-Private
subnet-0c714521bec793fea / EWB-B-Private
Details
Flow logs
Route table
Network ACL
CIDR reservations
Sharing
Tags
Related resources
Subnet ID
subnet-0c714521bec793fea
IPv4 CIDR
10.20.1.0/24
Availability Zone
euw1-az2 (eu-west-1a)
Default subnet
No
Customer-owned IPv4 pool
–
IPv6-only
No
DNS64
Disabled
Subnet ARN
arn:aws:ec2:eu-west-1:591697680381:subnet/subnet-0c714521bec793fea
Available IPv4 addresses
250
VPC
vpc-084569423cf72f99a | EWB-VPC-B
Auto-assign public IPv4 address
No
Outpost ID
–
Hostname type
IP name
Owner
591697680381
State
Available
IPv6 CIDR
–
Route table
rtb-013cc000659cd75f6 | EWB-B-Private-RT
Auto-assign IPv6 address
No
IPv4 CIDR reservations
–
Resource name DNS A record
Disabled
Block Public Access
Off
IPv6 CIDR association ID
–
Network ACL
acl-09281e71d5ec511e7
Auto-assign customer-owned IPv4 address
No
IPv6 CIDR reservations
–
Resource name DNS AAAA record
Disabled

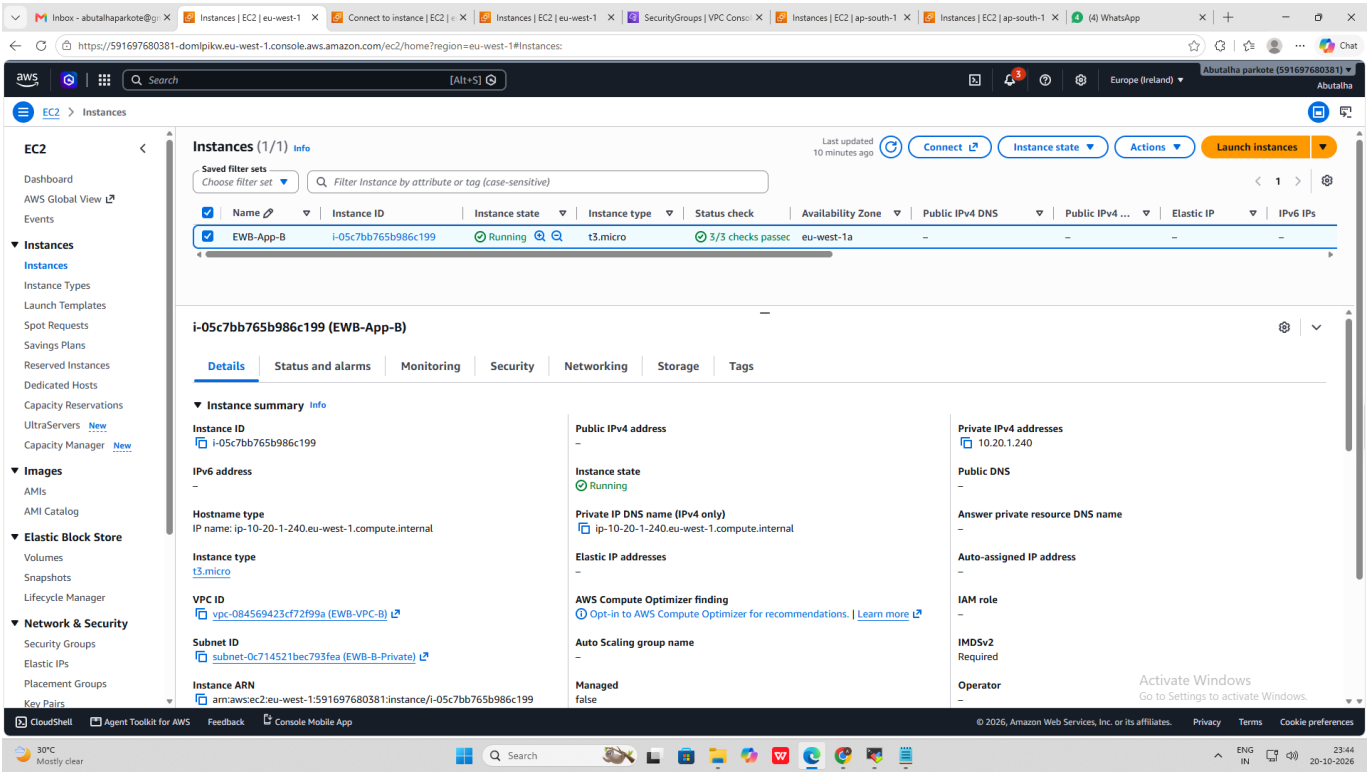

Route tables (1/4)
Create route table
EWB-A-PublicRT
rtb-0a72b8cb746bc0159 / EWB-A-PublicRT
Details
Routes
Subnet associations
Edge associations
Route propagation
Tags
Related resources
Route table ID
rtb-0a72b8cb746bc0159
VPC
vpc-006258cabde02bece | EWB-VPC-A
Main
No
Owner ID
591697680381
Explicit subnet associations
subnet-0b688aa29d8eb4ac3 / EWB--A-Public
Edge associations
–

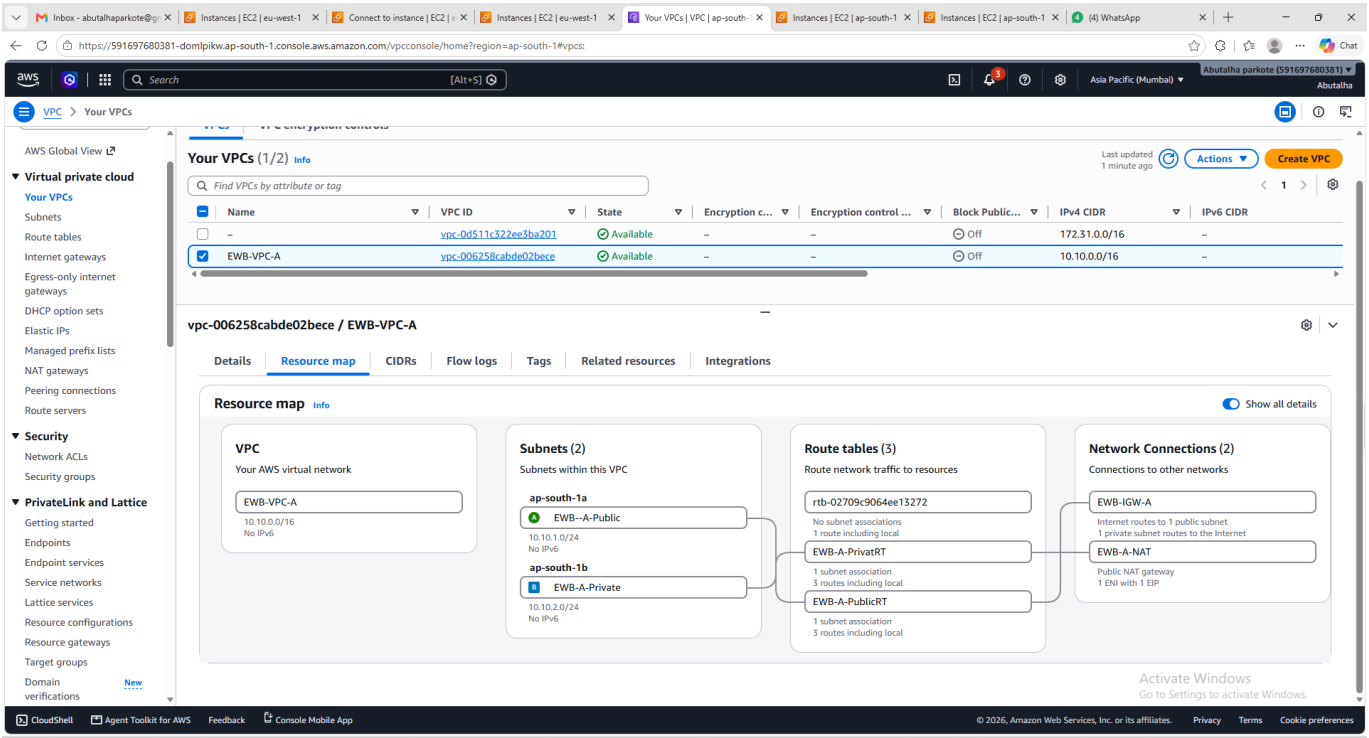

VPC > Route tables
Route tables (1/4) Info
Actions
Create route table
Name
Route table ID
Explicit subnet associ...
Edge associations
Main
VPC
Owner ID
rtb-02709c9064ee13272
rtb-008a7e5234c815d50
EWB-A-PrivatRT
rtb-04b9c6da7dbe12d08
EWB-A-PublicRT
rtb-0a72b8cb746bc0159
rtb-04b9c6da7dbe12d08 / EWB-A-PrivatRT
Details
Routes
Subnet associations
Edge associations
Route propagation
Tags
Related resources
Route table ID
rtb-04b9c6da7dbe12d08
Main
No
Explicit subnet associations
subnet-01251b67f33413101 / EWB-A-Private
Edge associations
VPC
vpc-006258cabde02bece | EWB-VPC-A
Owner ID
591697680381

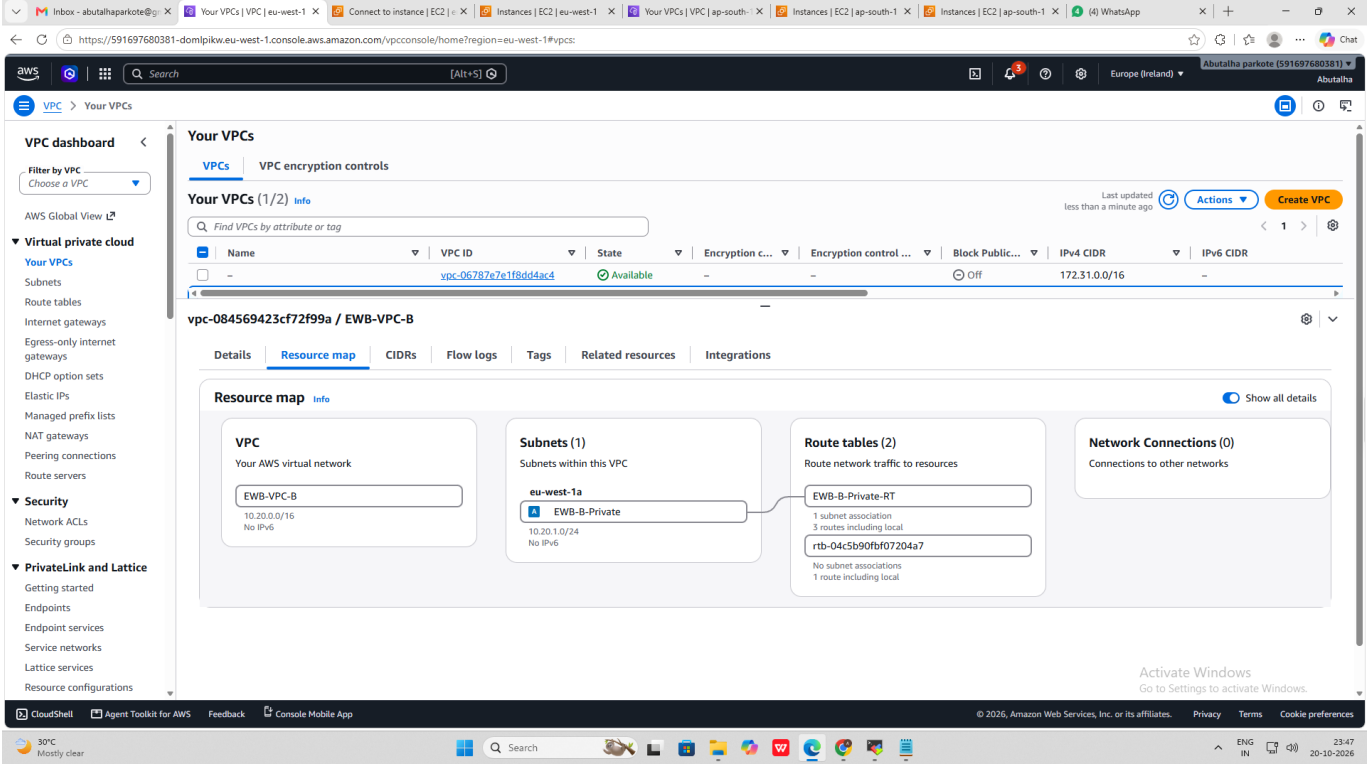

VPC > NAT gateways
NAT gateways (1/1) Info
Actions
Create NAT gateway
Name
NAT gateway ID
Connectivity...
State
State message
Availability ...
Route table ID
Primary public I...
Primary private I...
EWB-A-NAT
nat-077e268f658f40ced
Public
Available
Zonal
13.205.163.30
10.10.1.225
nat-077e268f658f40ced / EWB-A-NAT
Details
Secondary IPv4 addresses
Monitoring
Tags
NAT gateway ID
nat-077e268f658f40ced
Connectivity type
Public
State
Available
State message
NAT gateway ARN
arn:aws:ec2:ap-south-1:591697680381:natgateway/nat-077e268f658f40ced
Primary public IPv4 address
13.205.163.30
Primary private IPv4 address
10.10.1.225
Primary network interface ID
eni-055232b6ca282e6a0
VPC
vpc-006258cabde02bece / EWB-VPC-A
Subnet
subnet-0b688aa29d8eb4ac3 / EWB--A-Public
Created
Tuesday, October 20, 2026 at 22:57:06 GMT+5:30
Deleted

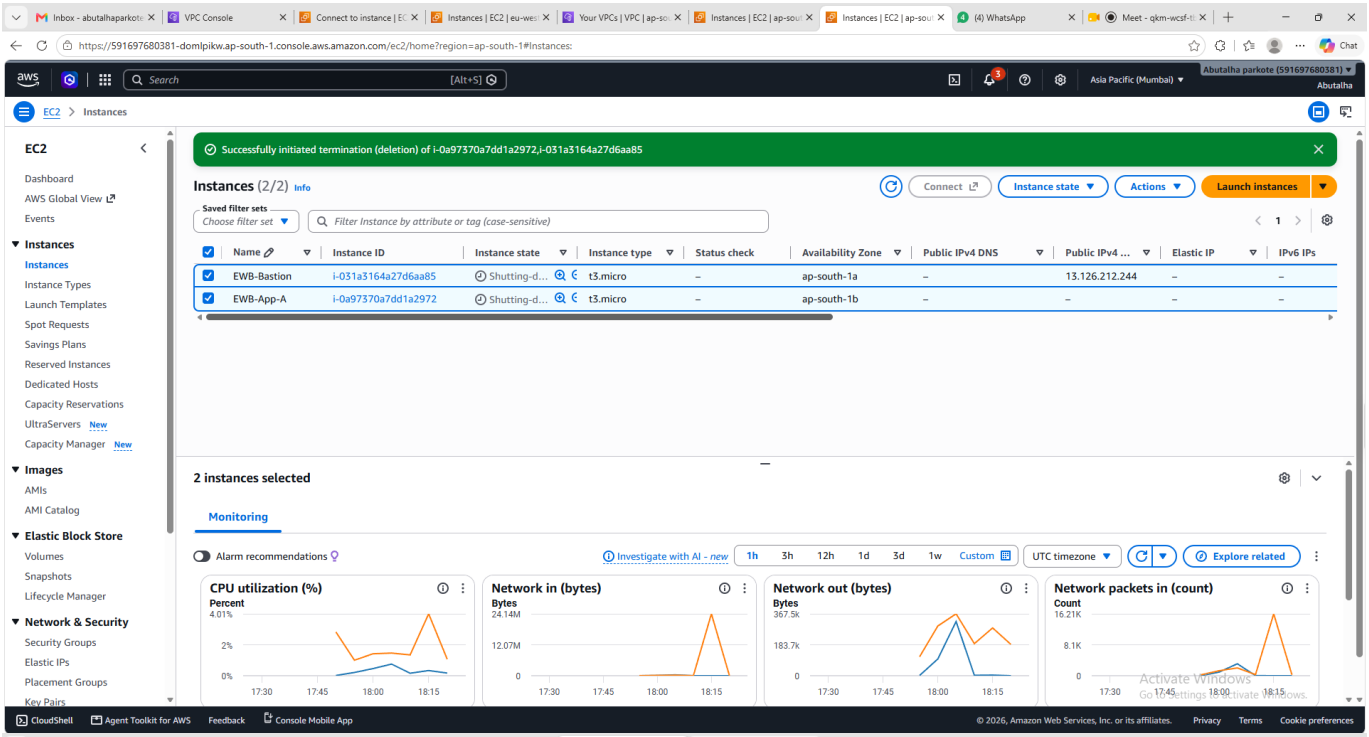

Peering connections (1/1)
Actions
Create peering connection
Name
Peering connection ID
Status
Requester VPC
Accepter VPC
Requester CIDRs
Accepter CIDRs
EWB-A-B-peer
pcx-00bbde40ea1696919
Active
vpc-006258cabde02bece / EWB...
vpc-084569423cf72f99a
10.10.0.0/16
10.20.0.0/16
pcx-00bbde40ea1696919 / EWB-A-B-peer
Details
DNS
Route tables
Tags
Requester owner ID
591697680381
Peering connection ID
pcx-00bbde40ea1696919
Status
Active
Expiration time
Accepter owner ID
591697680381
Requester VPC
vpc-006258cabde02bece / EWB-VPC-A
Requester CIDRs
10.10.0.0/16
Requester Region
Mumbai (ap-south-1)
VPC Peering connection ARN
arn:aws:ec2:ap-south-1:591697680381:vpc-peering-connection/pcx-00bbde40ea1696919
Accepter VPC
vpc-084569423cf72f99a
Accepter CIDRs
10.20.0.0/16
Accepter Region
Ireland (eu-west-1)

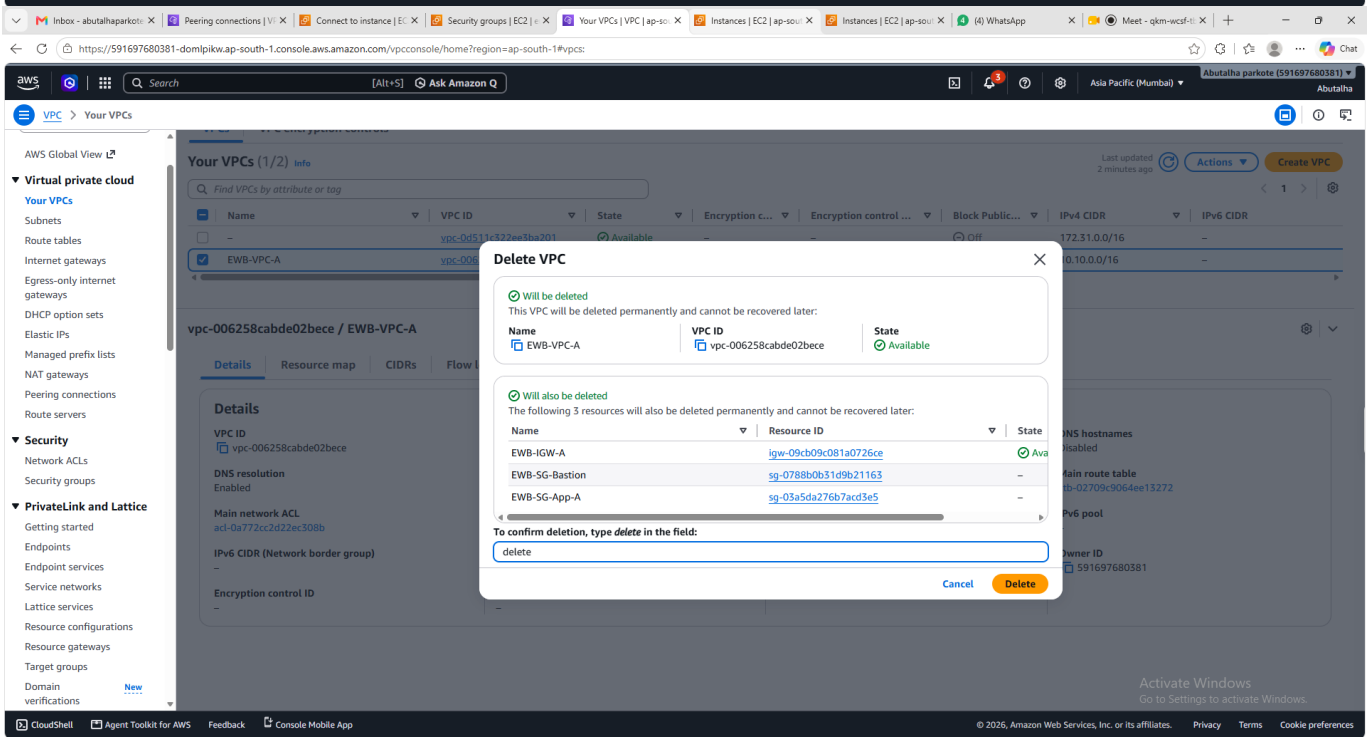

Inbound security group rules successfully modified on security group (sg-03a5da276b7acd3e5 | EWB-SG-App-A)
Security Groups (1/4)
Actions
Export security groups to CSV
Create security group
Name
Security group ID
Security group name
VPC ID
Description
Owner
sg-0da799ea82f295cef
default
vpc-006258cabde02bece
default VPC security group
591697680381
sg-0b9f9b48f22502da0
vpc-0d511c322ee3ba201
sg-0788b0b31d9b21163
EWB-A-SG
EWB-A-SG mumbai
sg-03a5da276b7acd3e5
EWB-SG-App-A
ewb-sg-app-a
sg-0788b0b31d9b21163 - EWB-A-SG
Details
Inbound rules
Outbound rules
Sharing
VPC associations
Related resources - new
Tags
Inbound rules (4)
Manage tags
Edit inbound rules
Security group rule ID
IP version
Type
Protocol
Port range
Source
sgr-0827bf9dcdaabf3b4
IPv4
All traffic
All
10.20.0.0/16
sgr-0628cd35949c7c381
sgr-0732052fea18b6f8c
SSH
TCP
22
0.0.0.0/0
sgr-0b6440f101ad6231c
All ICMP - IPv4
ICMP

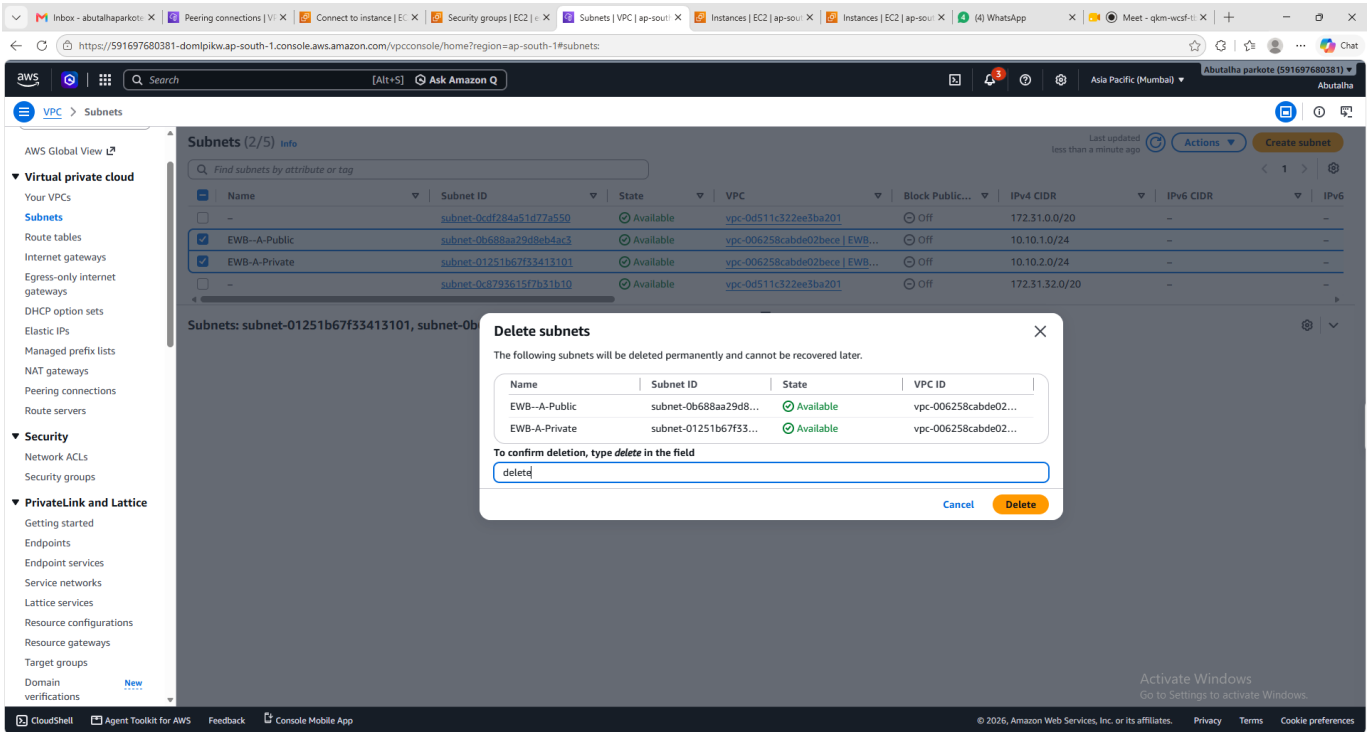

Instances (1/1) Info
Launch instances
EWB-App-B
i-05c7bb765b986c199
Running
t3.micro
eu-west-1a
i-05c7bb765b986c199 (EWB-App-B)
Details
Status and alarms
Monitoring
Security
Networking
Storage
Tags
Instance summary Info
Instance ID
i-05c7bb765b986c199
Public IPv4 address
Private IPv4 addresses
10.20.1.240
IPv6 address
Instance state
Running
Public DNS
Hostname type
IP name: ip-10-20-1-240.eu-west-1.compute.internal
Private IP DNS name (IPv4 only)
ip-10-20-1-240.eu-west-1.compute.internal
Answer private resource DNS name
Instance type
t3.micro
Elastic IP addresses
Auto-assigned IP address
VPC ID
vpc-084569423cf72f99a (EWB-VPC-B)
AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations. | Learn more
IAM role
Subnet ID
subnet-0c714521bec793fea (EWB-B-Private)
Auto Scaling group name
IMDSv2
Required
Instance ARN
arn:aws:ec2:eu-west-1:591697680381:instance/i-05c7bb765b986c199
Managed
false
Operator

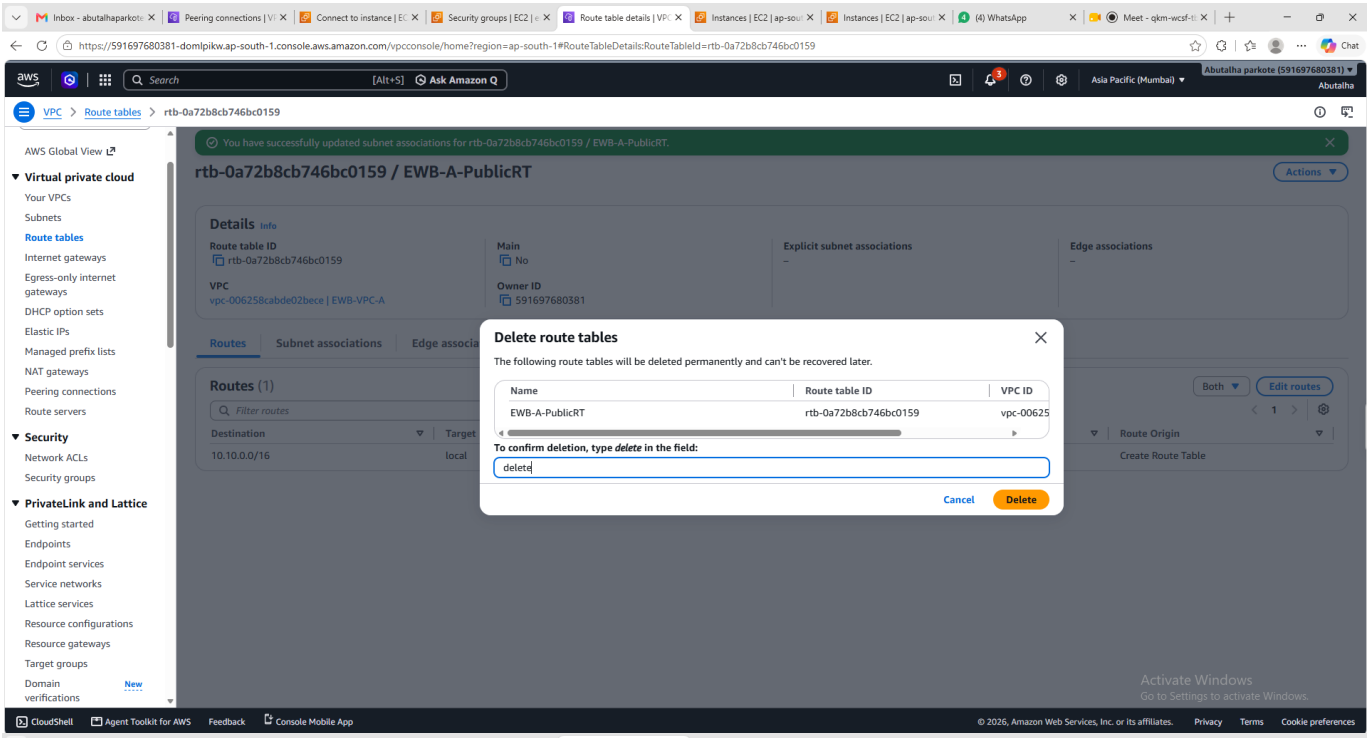

Your VPCs (1/2)
Name
VPC ID
State
EWB-VPC-A
vpc-006258cabde02bece
Available
10.10.0.0/16
vpc-006258cabde02bece / EWB-VPC-A
Details
Resource map
CIDRs
Flow logs
Tags
Related resources
Integrations
Resource map
VPC
Your AWS virtual network
EWB-VPC-A
10.10.0.0/16
No IPv6
Subnets (2)
Subnets within this VPC
ap-south-1a
EWB--A-Public
10.10.1.0/24
No IPv6
ap-south-1b
EWB-A-Private
10.10.2.0/24
No IPv6
Route tables (3)
Route network traffic to resources
rtb-02709c9064ee13272
No subnet associations
1 route including local
EWB-A-PrivatRT
1 subnet association
3 routes including local
EWB-A-PublicRT
1 subnet association
3 routes including local
Network Connections (2)
Connections to other networks
EWB-IGW-A
Internet routes to 1 public subnet
1 private subnet routes to the Internet
EWB-A-NAT
Public NAT gateway
1 ENI with 1 EIP

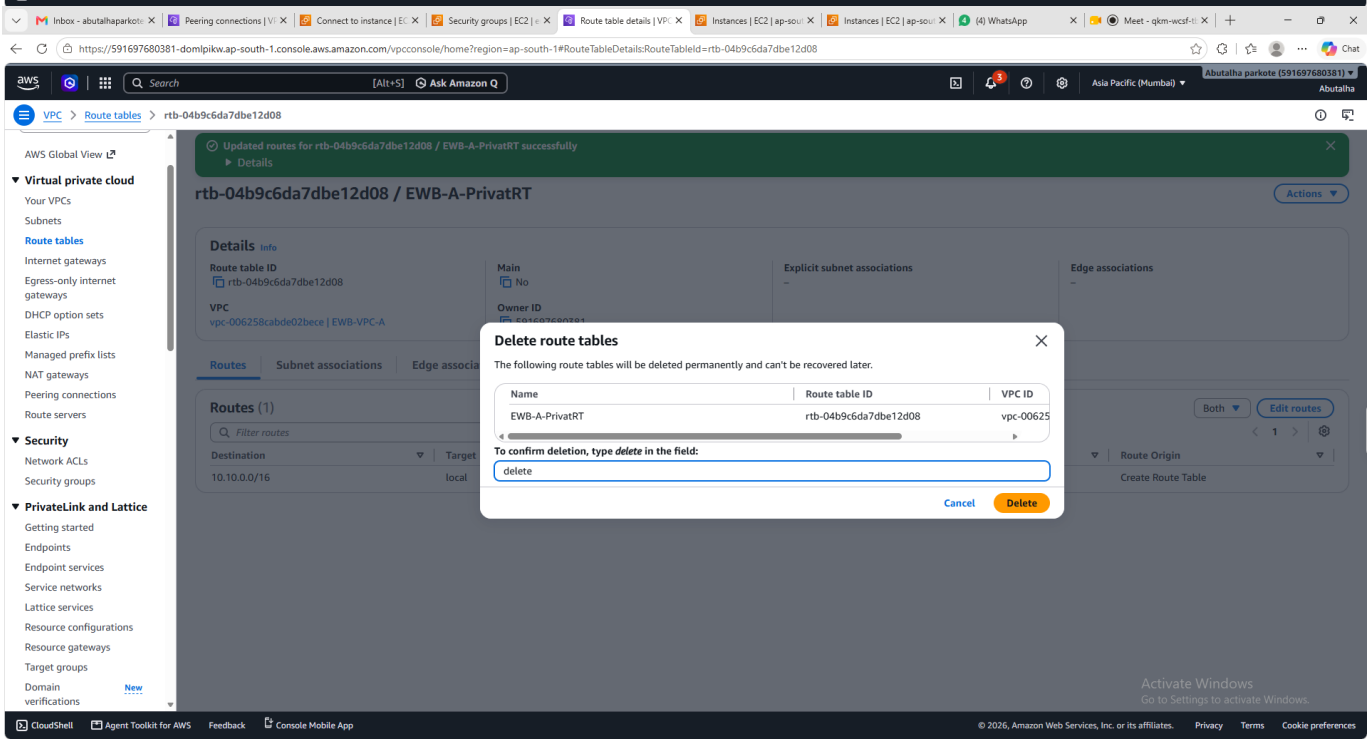

Your VPCs
VPCs
VPC encryption controls
Your VPCs (1/2)
Name
VPC ID
State
vpc-06787e7e1f8dd4ac4
Available
172.31.0.0/16
vpc-084569423cf72f99a / EWB-VPC-B
Details
Resource map
CIDRs
Flow logs
Tags
Related resources
Integrations
Resource map
VPC
Your AWS virtual network
EWB-VPC-B
10.20.0.0/16
No IPv6
Subnets (1)
Subnets within this VPC
eu-west-1a
EWB-B-Private
10.20.1.0/24
No IPv6
Route tables (2)
Route network traffic to resources
EWB-B-Private-RT
1 subnet association
3 routes including local
rtb-04c5b90fbf07204a7
No subnet associations
1 route including local
Network Connections (0)
Connections to other networks

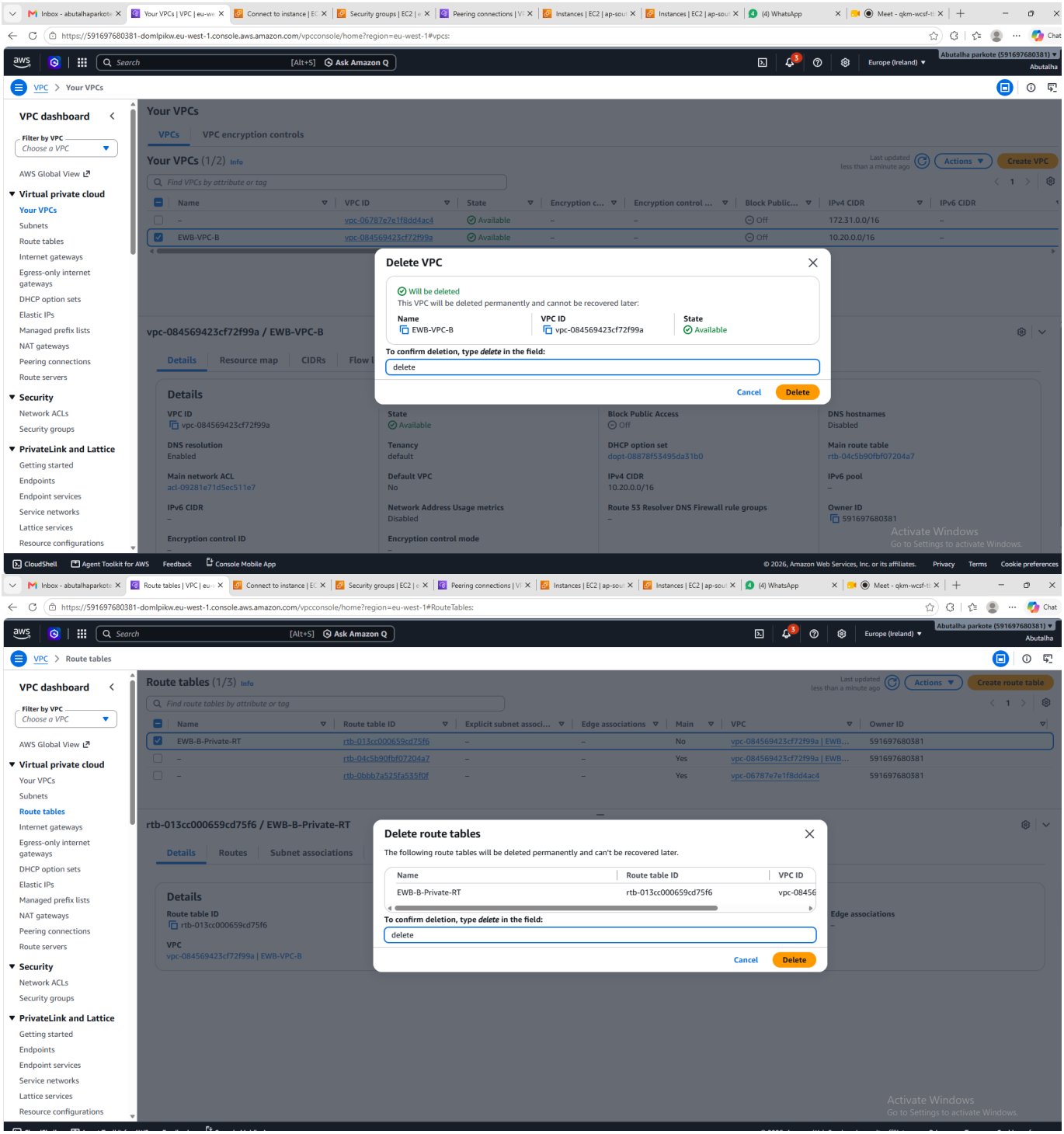

Successfully initiated termination (deletion) of i-0a97370a7dd1a2972,i-031a3164a27d6aa85
Instances (2/2)
EWB-Bastion
EWB-App-A
2 instances selected
Monitoring
CPU utilization (%)
Network in (bytes)
Network out (bytes)
Network packets in (count)

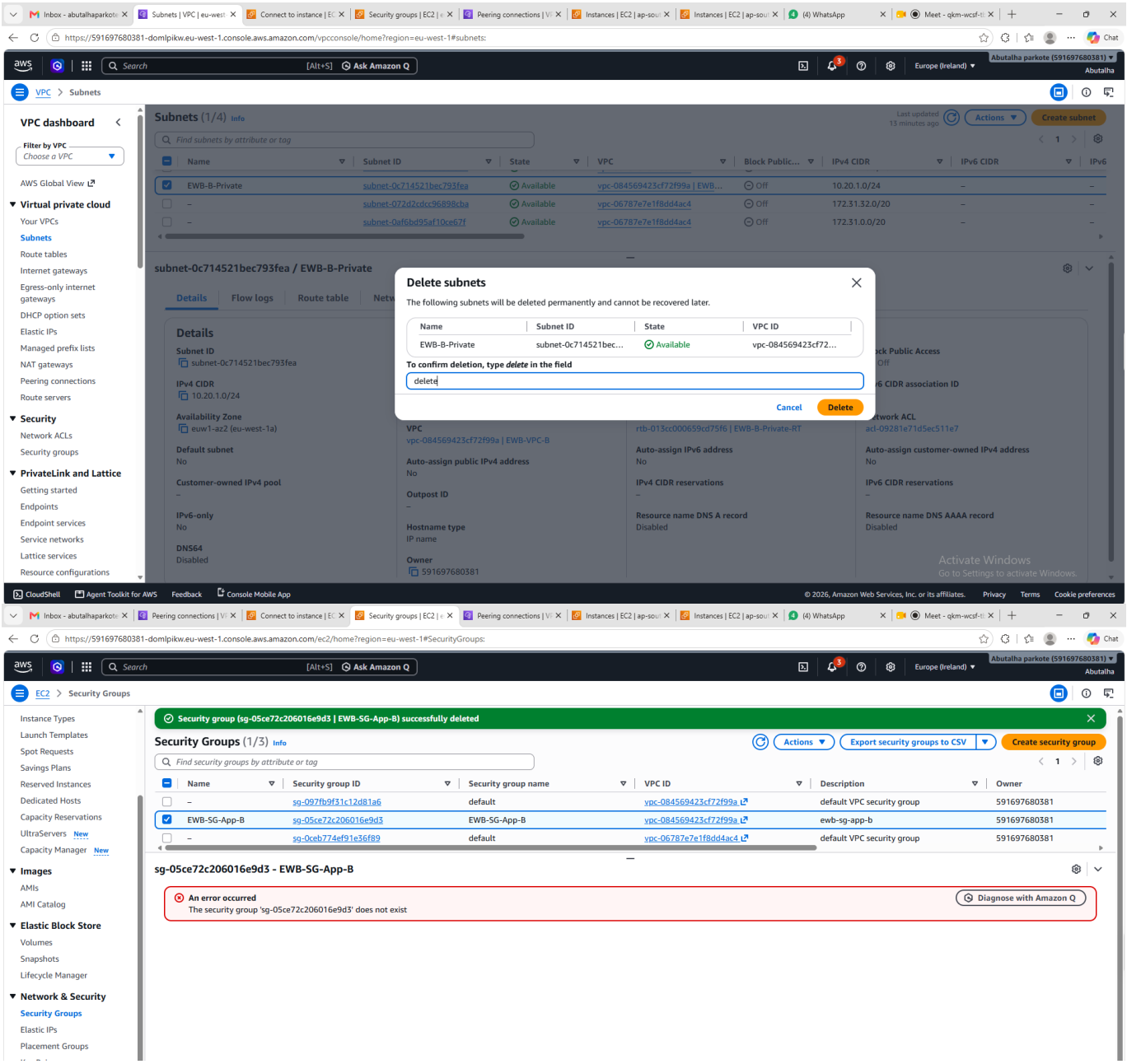

Your VPCs (1/2)
EWB-VPC-A
Delete VPC
Will be deleted
This VPC will be deleted permanently and cannot be recovered later:
Will also be deleted
The following 3 resources will also be deleted permanently and cannot be recovered later:
EWB-IGW-A
EWB-SG-Bastion
EWB-SG-App-A
To confirm deletion, type delete in the field:
delete
Cancel
Delete

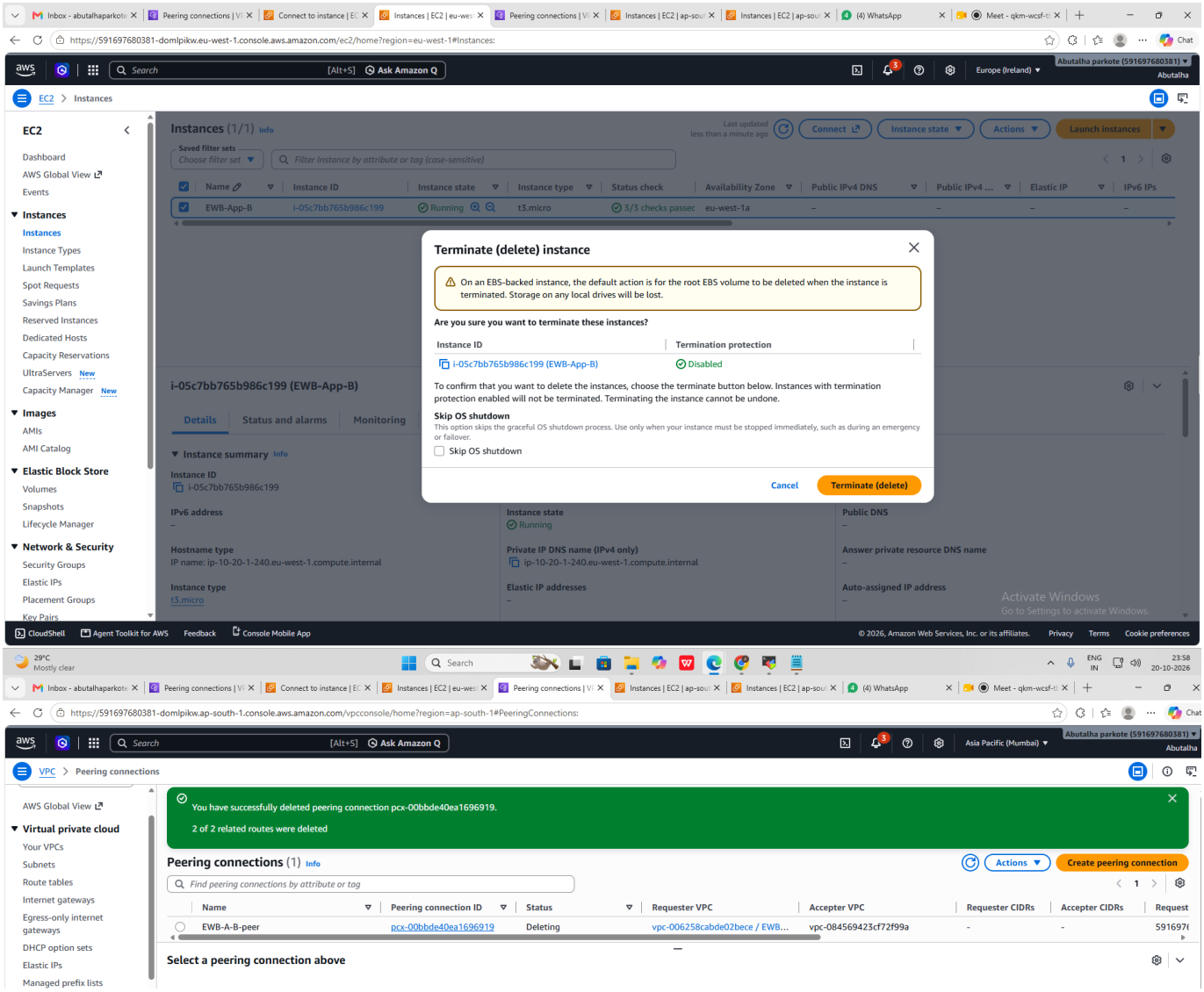

Subnets (2/5)
Delete subnets
The following subnets will be deleted permanently and cannot be recovered later.
Name
Subnet ID
State
VPC ID
EWB--A-Public
subnet-0b688aa29d8...
Available
vpc-006258cabde02...
EWB-A-Private
subnet-01251b67f33...
Available
vpc-006258cabde02...
To confirm deletion, type delete in the field
delete
Cancel
Delete


rtb-0a72b8cb746bc0159 / EWB-A-PublicRT
Delete route tables
The following route tables will be deleted permanently and can't be recovered later.
Name
Route table ID
VPC ID
EWB-A-PublicRT
rtb-0a72b8cb746bc0159
vpc-00625
To confirm deletion, type delete in the field:
delete
Cancel
Delete


rtb-04b9c6da7dbe12d08 / EWB-A-PrivatRT
Delete route tables
The following route tables will be deleted permanently and can't be recovered later.
Name
Route table ID
VPC ID
EWB-A-PrivatRT
rtb-04b9c6da7dbe12d08
vpc-00625
To confirm deletion, type delete in the field:
delete
Cancel
Delete


Delete VPC
Will be deleted
This VPC will be deleted permanently and cannot be recovered later:
Name
EWB-VPC-B
VPC ID
vpc-084569423cf72f99a
State
Available
To confirm deletion, type delete in the field:
delete
Cancel
Delete
Delete route tables
The following route tables will be deleted permanently and can't be recovered later.
Name
Route table ID
VPC ID
EWB-B-Private-RT
rtb-013cc000659cd75f6
To confirm deletion, type delete in the field:
delete
Cancel
Delete


Subnets (1/4)
Delete subnets
The following subnets will be deleted permanently and cannot be recovered later.
Name
Subnet ID
State
VPC ID
EWB-B-Private
subnet-0c714521bec...
Available
vpc-084569423cf72...
To confirm deletion, type delete in the field
delete
Cancel
Delete
Security group (sg-05ce72c206016e9d3 | EWB-SG-App-B) successfully deleted
Security Groups (1/3)
An error occurred
The security group 'sg-05ce72c206016e9d3' does not exist
Diagnose with Amazon Q



10. Challenges and Learning Outcomes

Challenge encountered and root cause

I faced issues verifying internet connectivity and resolving communication problems between the two VPCs. The main challenge was identifying the routing and configuration issues affecting the peering connection.

Troubleshooting steps

I checked the VPC peering status, route tables, subnet associations, and security group rules. I verified that the correct inbound rules were configured to allow the required SSH and ICMP traffic.

What I learned about VPC peering and route tables

I learned how VPC Peering enables private communication between two VPCs and how route tables must contain the correct routes for both forward and return traffic.

What I learned about SSH and security groups

I learned how to access private EC2 instances through a Bastion Host using SSH and how security group inbound rules to control network access.

Cost-control and cleanup lessons

I learned the importance of terminating unused EC2 instances, deleting unnecessary networking resources, and removing unused Elastic IPs and NAT Gateways to avoid unexpected AWS charges.

11. Conclusion

This AWS Cloud Internship project provides practical experience in designing and configuring secure communication between two Amazon VPCs using VPC Peering. I learned how to create VPCs and subnets, configure route tables, establish a peering connection, and manage security group rules for controlled network access.

The project also helped me understand how to access private EC2 instances through a Bastion Host using SSH and how to troubleshoot connectivity issues by checking internet access, routing configurations, and inbound rules. Through ping tests and controlled ICMP failure and recovery, I gained a better understanding of network connectivity and security.

Overall, this project strengthened my practical knowledge of AWS networking, network security, and troubleshooting. It also highlighted the importance of validating configurations, following security best practices, and cleaning up unused resources to control AWS costs.

Appendix A. Screenshot Evidence Checklist

- Final architecture diagram.
- VPC details for both VPCs showing IDs and CIDRs.
- Subnet details showing subnet IDs, CIDRs, and Availability Zones.
- All three route tables, including routes and subnet associations.
- Active peering connection with requester/accepter VPC details.
- Inbound rules for the Bastion, App-A, and App-B security groups.
- EC2 instance details showing private IPs and no public IP on private hosts.
- Successful baseline ping and SSH evidence.
- Controlled ICMP failure and successful recovery evidence.
- Cleanup evidence and final resource state.

End of Report